Smoking Gun? Linking Gun Ownership to Neighborhood Crime

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Abstract

Using linked individual data on concealed handgun permits (CHP), reported crimes and arrests, I examine the dynamics of gun-ownership, victimization and neighborhood crime. I initially show that new CHP holders are 68% more likely to be crime victims with the largest increase due to having a firearm stolen. I then estimate the impact of guns on neighborhood crime through a shift-share instrument based on spikes in gun sales after high-profile events that impact beliefs about future gun rights and existing variation in a neighborhood's share of likely future gun-owners (Republicans). Results show a 5.3% increase in neighborhood crime for the new guns associated with each additional CHP issued. Gun-related crimes suggest a role for guns stolen and moved to the illegal gun market as well as increased use of guns for violent crimes.

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1. Introduction

The tragedy of gun violence is revisited with great regularity and each high profile shooting renews debates about the role of guns in US society. Even after shooting events such as Columbine High School in 1999, Rep. Gabrielle Giffords in 2011, Sandy Hook Elementary School in 2012, San Bernardino in 2015 and Parkland High School in 2018, national gun legislation was left unchanged and typically led to massive increases in gun purchases.[2] Local gun laws and enforcement did react to these high profile events with a number of subsequent changes (both more and less restrictive) to gun accessibility and permitting (Yourish et al. (2013)). These local policy changes have been incorporated into recent research which consistently finds a positive relationship between gun availability and crime.[3] Unfortunately, how gun policy and availability translates into criminal outcomes is less clear and complicated by limited data on gun ownership as well as a diverse set of channels through which legal guns end up in the hands of criminals.

At the crux of this policy debate is how are legal guns obtained and used by criminals? As highlighted in several papers (Cook (2018) , Cook et al. (2014) , Cook et al. (2015a) , Collins et al. (2017) , Cook and Braga (2001)), we have evidence to support a number of mechanisms through which criminals obtain guns. Specifically, gun tracing data support a role for interstate trafficking or larger scale straw purchasing within jurisdictions that have limited supply and typically more regulated gun access (Cook et al. (2015a) , Collins et al. (2017)).[4] Research supports a substantial role for informal and local social connections to transfer guns between legal and illegal purchasers (Braga et al. (2010) , Papachristos (2009) , Papachristos et al. (2015) , Papachristos and Wildeman (2014) , Miller et al. (2017)) as well as evidence of a robust market for individuals to steal guns from homes, cars and persons and sell to criminals (Kleck and Wang (2008) , Khalil (2017)). Finally, there is even reason to suspect that licensed gun dealers may be providing guns to individuals that later use them in criminal acts (Cook (2018), Cook et al. (2015a)).

Even with evidence supporting a diverse set of channels through which criminals obtain guns, there are some drawbacks to this literature. First, the decision to obtain a legal gun as well as how it is stored and used is endogenous to both individual as well as neighborhood crime risk. This endogeniety may be reflected in both the use of a legal gun as well as how a gun may enter the illegal market as underlying crime risk impacts

[2] Luca et al. (2020) provides evidence that mass shootings lead to a 15% increase in gun legislative discussions but no actual changes in the law. Levine and McKnight (2017) find approximately a 60% increase in gun purchases after Sandy Hook.

[3] Cook and Donohue (2017) and Cook and Ludwig (2019) provide a review of this literature.

[4] A straw purchaser is defined as anyone who legally obtains a gun on behalf of another individual who is unable to obtain the gun legally.

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both the demand and supply of guns.[5] Second, this literature is often based on surveys of inmates, small samples of recovered guns or aggregated measures of gun availability and crime. Given data scarcity, there is limited revealed preference data on guns and crime at the individual or even neighborhood level. This type of data may help disentangle some of the various findings in the literature. For example, if individual gun owners are later arrested for crimes or become victims of crime, it reveals quite different dynamics of how legal guns are being obtained for criminal acts. Disaggregated details about crimes also reveal how guns may enter illegal networks as one can use the location and timing of events like a stolen gun and look at subsequent crimes using guns.

The paper presented here provides two main contributions to understanding how local gun ownership impacts crime. First, I provide evidence on criminal victimization and arrests for individuals who obtain concealed handgun permits (CHPs). This evidence is compelling since my dataset allows for a rich description of the timing, location and details of victimization and arrests and thus can leverage temporal variation at the individual level while controlling for a rich set of confounding neighborhood factors. Individual analysis is rare in the guns and crime literature more broadly and is a yet an unexplored determinant of crime victimization. Second, I examine how neighborhood gun ownership impacts crime with a focus on crimes that involved a gun. In order to provide causal estimates, I formalize and expand existing literature that uses national shocks to gun demand due to school shootings or other national events (Levine and McKnight (2017), Depetris-Chauvin (2015) , Koenig and Schindler (2019)) by introducing a shift share instrument.

My analysis begins with the creation of a unique individual panel dataset that links registered voters to concealed handgun permits (CHPs) and reported crime victims from 2007 through 2011 in Charlotte, NC.[6] All of these datasets are linkable by full name, birth year and residential address. CHP holders are typically white, age 30 and older, male and politically more Republican. In order to limit the influence of unobserved crime risk, I begin my analysis of crime victimizations by doing an exact match between CHP and non-CHP holders based on voter attributes, voter victimization history, neighborhood of residence and the month that a new CHP is issued and estimate a difference-in-difference model. Since being a previous crime victim is the best measure of crime risk, my preferred models will further limit analysis to only individuals without a criminal victimization. Individual analysis highlights that getting a CHP increases overall victimization by

[5] For example, higher crime in my neighborhood may lead more criminals to demand guns or obtain guns through theft while at the same time inducing an increase in the supply of legal guns obtained for personal protection.

[6] Approximately 86% of CHP holders are also registered voters making them representative of the main population of potential gun owners. In comparison, about 75% of the 18+ population in Charlotte are registered voters.

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68%, but effects are limited to non-violent crimes. Gun-specific crimes highlight a 228% increase in stolen gun victimization following a new CHP.

To estimate the relationship between gun ownership and neighborhood crime, I leverage idiosyncratic increases in individual gun ownership that coincide with nationwide concerns about future gun rights following the inauguration of Barack Obama and the shooting of Rep. Gabrielle Giffords.[7] I adopt a shift share instrument based on share Republican in a given Census Block Group (CBG) neighborhood and national flows of gun background checks to vary new concealed handgun permits (CHPs) for neighborhoods in Charlotte, NC.[8] This approach avoids some of the methodological shortcomings of existing papers based on more aggregate panel analysis using gun restriction policies since it allows plausibly exogenous shocks to concerns about gun availability to vary the stock of legal guns. Neighborhood analysis also incorporates spillover effects from individual gun owners to the neighborhood which is an important component of the larger social costs of guns.

A number of specifications highlight the validity of the shift share instrument. Following a recent literature that provides two perspectives on valid shift share instruments, I provide evidence focusing on exogenous shifters over time (e.g. Borusyak et al. (2018)) and that initial shares are conditionally exogenous (e.g. Goldsmith-Pinkham et al. (2018)). The uptick in gun purchases following the inaguration of Barack Obama and the shooting of Rep. Gabrielle Giffords are unlikely to be related to local crime trends in Charlotte, but they may relate to formal or informal changes in local gun policies (there is not clear evidence of specific law changes, but law enforcement may change). To avoid this concern, my main instrument is based on NICS (National Instant Criminal Background Check System) background checks for gun purchases outside of North Carolina and I show results are robust to different variants of this instrument including omitting neighboring states.[9] I also provide evidence that the initial shares of Republicans are conditionally exogenous by showing that once I control for other neighborhood demographic information, there are minimal differences in crime levels

[7] Google trend analysis will highlight increased beliefs about national gun legislation consistent with the timing of specific events. I do not limit my analysis to just the months prior to and after the inauguration of Obama or shooting of Giffords to capture general concerns about gun rights that changed and evolved during and after these events. I later show that results to be robust to just focusing on the 6 months prior to and after the inaugration of Obama.

[8] I later provide a full discussion and analysis of the scale upon which to measure neighborhoods, but initially justify CBGs based on a sizeable literature in criminology that has measured the distance between a arrestee's home address and the location of reported crimes. Across the following studies Phillips (1980) , Gabor and Gottheil (1984) , Rhodes and Conly (1981) , Repetto (1974) , Barker (2000) and Capone and Nichols Jr (1976) , criminals traveled on average between 0.35 and 2.41 miles across a range of urban settings, which is about the size of a CBG in my data.

[9] See Knight (2013) for evidence that illegal guns disproportionately originate from neighboring states with weaker guns laws.

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between Republican and non-Republican neighborhoods as well as similar pre-trends in crime outcomes. I further highlight that Republican and non-Republican neighborhoods exhibit similar trends in home prices and the income of new residents over my study period.

Using my shift share instrument, results for neighborhood crime highlight a consistent positive impact of an additional neighborhood CHP on total (5.3%) and violent neighborhood crimes (4.8%).[10] Immediate effects (up to 2 months) are larger for most crime outcomes, but persist for some categories of gun-related crimes. Highlighting a more direct role of guns in criminal outcomes, I estimate models on three hypothesized effect of more legal guns in a neighborhood: 1) guns increase violent encounters between individuals leading to greater injuries (16% increase in crimes with serious injury); 2) guns increases the payoff from property crimes (20% increase in crimes where a gun was reported stolen); 3) legal guns end up in illegal markets and are more likely to become part of violent crimes in the future (32% increase in violent crimes using a gun).

Subsequent analysis that examines specific crime categories, as well as incorporates a sample of The Trace data for Charlotte suggests a limited role of interstate trade in guns or large-scale arms dealers, but rather a narrative that property crimes are yielding more guns as CHPs increase.[11] These stolen guns coincide with a range of criminal activity in the neighborhood consistent with gang activity, drug trafficking or providing illegal weapons to criminals. I am unable to rule out criminals acquiring guns through social networks or illegal dealers, but do find evidence that CHP holders are less likely to be subject to arrest after acquiring a CHP. This fact could suggest the presence of straw purchasers who need to avoid criminal detection, but the fact that these individuals are not being arrested for weapons crimes does limit the scale of this underlying channel as being the main driver of increases in neighborhood crime.

Understanding the role of legal guns in neighborhood crime is also important in light of recent literature (e.g. Chetty and Hendren (2018), Chetty et al. (2016) , Ludwig et al. (2013)) that highlights the role of place in the long-term outcomes of children. If guns positively impact crime (which negatively impacts neighborhoods more broadly), guns may exacerbate the concentration of disadvantage as mobile, likely better off individuals move away from high crime neighborhoods.[12] Furthermore, the impact of guns on crime

[10] Effect sizes can vary based on the scaling of CHPs and crime, but my main estimates are consistent with the marginal impact of adding the average amount of new neighborhood CHPs for 2.5 months (0.4 new neighborhood CHPs per month) on 6 months of crime outcomes. Each additional CHP coincides with multiple guns (based on Brauer et al. (2013)), so effects could be interpreted as about 4.4 guns leading to these estimated impacts.

[11] The Trace data collects administrative police data which provides individual observations linking original purchasing jurisdiction to where the gun is later recovered as part of a criminal investigation. This data is available for download at https://www.thetrace.org/missing-pieces-data/ .

[12] Cullen and Levitt (1999) highlight the role of crime in residents leaving the urban core for suburban

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specifically is important given the large potential for social spillovers onto non-gun owners in terms of neighborhood crime victimization. To put my results in terms of actual costs, I take estimates on the costs of crime from McCollister et al. (2010) and estimate crime category specific models to generate a per capita social cost of crime. Each additional neighborhood CHP and its associated guns generate a social cost of crime of $76.55 per person.

In terms of policy implications, the localized nature of impacts indicates that policies such as Child Access Prevention and gun storage laws that safeguard guns may limit their theft and thus prevent future neighborhood violent crimes. Even more promising for thwarting this mechanism of stolen guns are biometric recognition devices that limit gun use to only registered owners ((Cook and Goss 2014)). These policy implications are not in contradiction to existing findings of a larger role of straw purchasers and interstate commerce for guns in places like Chicago (Cook et al. (2015a) , Collins et al. (2017)) since the regulatory environment and supply of guns in southern cities like Charlotte may be more conducive to a local supply of guns through theft.[13] In the end, my conclusions are specific to CHP holders and thus may not reflect the specific impacts of other legal handgun purchase such as through pistol purchase permits (PPP) or gun shows.[14]

The rest of the paper continues with a discussion of related literature in Section 2 and details about gun permitting in North Carolina and my dataset in Sections 3 and 4. Section 5 provides individual analysis of the impact of gun ownership on crime victimization and arrest. Sections 6 and 7 describe the shift share instrument and provide results for the link between gun ownership and neighborhood crime. Finally, Section 8 highlights mechanisms behind my results and the implied social costs of guns on crime from neighborhood estimates.

2. Related Literature

This paper contributes to a larger literature on the social costs of gun ownership. This literature does provide consistent evidence that more guns or easier access to guns in the home lead to higher rates of accidental shootings (e.g. Levine and McKnight (2017) , Grossman et al. (2012), DeSimone et al. (2013) ,Wiebe (2003a) , Hemenway (2011) , Morrow and Hudson (1986)) and higher rates of suicide (e.g. Anestis and Houtsma

neighborhoods.

[13] Collins et al. (2017) does find stronger support for the stolen gun channel in New Orleans and Prince George's County, MD based on inmate interviews.

[14] The fact that I do later provide evidence that the timing of background checks as well as the political affiliation of neighborhoods with PPP holders is quite similar to CHP holders is suggestive that these effects may hold across other types of gun purchases.

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(2018) , Miller et al. (2006), Miller et al. (2007), Ghiani et al. (2019) and Wiebe (2003b)). The debate over the impact of legal guns on crime has a more contentious history with papers by Lott and Mustard (1997) and Moody (2001) finding that more guns lower violent crimes, while a parallel literature found positive and no effects of guns on crime (Ludwig (1998) , Black and Nagin (1998) and Donohue and Ayres (2003)). These papers focus on Right To Carry (RTC) laws and recent work (Donohue et al. (2019) and McElroy and Wang (2017)) provides more conclusive panel evidence that less restrictive RTC handgun laws increase violent crime.[15] Most of the concerns or controversy in this earlier literature are likely a result of endogenous policy changes or violations of parallel trends assumptions across RTC policies.

There are also a number of papers that focus on non-RTC variation in gun access. Duggan (2001) provides evidence that upticks in gun magazine subscriptions and gun shows precede increases in gun homicides but not non-gun homicides with limited impacts on other crimes. Castle Doctrine or "stand your ground" laws allow the use of guns to defend one's home or property and both Cheng and Hoekstra (2013) and McClellan and Tekin (2017) find that these laws lead to a significant increase in homicides. A number of papers (Lang (2013), Williams Jr (2017) and Duggan et al. (2011)) focus on the removal or addition of restrictions (e.g. background checks, purchase permits, etc.) on submarkets (e.g. private gun sales, gun shows, etc.) for gun sales and find a mix of either no impact or increases in violent crimes and the context seems to matter. Tannenbaum (2020) uses the release of information about the residential location of Concealed Handgun Permit (CHP) holders and finds no effect of the public disclosure of a CHP on crime at that address. Across this literature, lowering the psychological (e.g. "stand your ground laws") or transaction costs (e.g. permits, background checks) of obtaining a gun appears to increase gun related violence.

Estimating the relationship between guns and crime victimization contributes to a growing literature on the determinants of crime victimization. This literature spans a number of contexts but provides evidence of how policies, private actions and individual attributes impact criminal victimizations. For example, Ayres and Levitt (1998) and Gonzalez-Navarro (2013) show that the adoption of Lojack anti-auto theft devices deterred auto theft more broadly. Vollaard and Van Ours (2011) show that a policy to require burglary proof windows and doors for new homes in the Netherlands decreased burglaries by 26% and Van Ours and Vollaard (2016) show that a European Union policy that required

[15] Right to Carry (RTC) are state-level laws that determine whether a permit is required and who is eligible to carry a concealed firearm on their person. In "unrestricted" states, the legality of carrying a concealed firearm is unrestricted; whereas "shall issue" policies are more restrictive and government officials shall provide permits to all eligible residents. "may issue" policies provide discretion to local officials in terms of permitting and in practice a state might prohibit any resident from carrying a concealed firearm in public but currently no states do this.

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automatic engine immobilizers decreases auto theft by 40%.

Actions by individual may directly contribute to higher and lower victimization rates. Specifically, Chalfin et al. (2019) show that as an individual ages into legal drinking in the US at 21, victimization rates increase for both property and violent crimes. Randi Hjalmarsson and Mitrut (2021) finds a similar result using policy changes regarding legal alcohol consumption in Netherlands. More broadly, papers (Jensen and Brownfield (1986), Mayhew and Elliott (1990) ,Sampson and Lauritsen (1990), Wittebrood and Nieuwbeerta (1999), Pedersen (2001)) in criminal justice show that routine activity or life style theory explains the positive association between being both a victim and offender.

Beyond simply predicting who becomes a crime victim, scholars have also focused on the individual impacts of crime victimization. For example, Braakmann (2012) show that higher victimization risks in Mexico lead to sleep deprivation and individuals taking measures to protect their homes. Bindler and Ketel (2019) show that victimization leads to longer-term negative effects on earnings and short term negative impacts on health. Mahuteau and Zhu (2016) find a negative impact of physical violence and property crimes on subjective well-being in Australia. Cornaglia et al. (2014) show an interconnection between violent crime and mental health with a decrease in mental well-being after violent crime victimization and that the violent crime rate has a negative impact on mental well-being of nonvictims. Finally, Currie et al. (2018) show negative impacts on infant health when a pregnant women is subject to interpersonal violence.

3. Background on Gun Permitting

Before providing details on data and estimation, one needs to understand how individuals obtain handguns in NC and the role of gun-permitting in generating my main measure of gun ownership. In 1995, North Carolina changed from a "no issue" to a "shall issue" state, which meant that prior to 1996, concealed handguns were not allowed and starting in 1996, residents could conceal carry if issued a permit by the state. North Carolina also requires a pistol purchase permit (PPP) to buy a handgun and NC Gen. Stat. 14-402 states that all sales of handguns (private or licensed gun dealers) must only be to permitted buyers. Specifically, a Concealed Handgun Permit (CHP) is needed to be able to carry a handgun under clothing or in other ways concealed into almost any structure or outside. There are some restrictions on carrying guns in government buildings as well as institutions that serve alcohol, but the latter has been allowed in subsequent legislative sessions. A person must be 21 years old, a resident of North Carolina for at least 30 days, have a valid driver's license, undergo a mental health medical background

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check (this requirement was removed in 2015), undergo a criminal background check and participate in a firearm safety course.

The background check does provide some discretion since a CHP may be approved or denied based on "good moral character" which is loosely defined as "honesty, fairness, and respect for the rights of others and for the laws of the State and nation" (Supreme Court of NC. Willis, 288 N.C. 1,10 1975). There is some evidence of individuals with previous arrests receiving CHPs and based on my analysis, about 1.8% of new CHP holders between 2007 and 2011 had at least one previous arrest and 0.3% had a felony arrest in the previous 5 years.[16] In practice, only a felony conviction would disqualify an individual from obtaining a CHP. An application for a CHP costs $90 and allows a person to purchase unlimited guns without any additional background checks. The process for obtaining a CHP typically takes about 45 days, but can be longer or shorter depending on the number of applications.

The choice of a CHP versus simply purchasing a gun (pistol puchase permit (PPP)) is based on two main differences between these two types of ownership: 1) A CHP allows for multiple gun purchases for up to five years while a PPP allows for only one gun to be purchased. 2) A CHP allows for concealed carry while a PPP is more limited in the usage and carrying options for prospective gun owners. The exact reasons for choosing a CHP versus a PPP are not well documented, but CHPs are more amendable to individuals that want to regularly carry a gun outside the home for protection.[17] Alternatively, one could purchase a handgun from a private owner or gun show, or one could purchase a long gun which does not require a permit. These alternative methods of gun ownership are universally difficult to measure.

4. Data

In order to examine the impact of the supply of legal guns on individual crime victimization, one requires rich and spatially disaggregate data on reported crimes, arrests and permitted gun ownership. The requirements of such data limits national analysis but does allow for specific jurisdictions that have publicly available reported crime data as well as concealed handgun or purchase permit microdata available. In my case, the city of Charlotte and specifically Mecklenburg County which encompass the city of Charlotte, NC plus a few smaller towns has made this data available. Specifically, I incorporate

[16] For the general population of voters, 8.3% had any arrests and 0.8% had a felony arrest in the previous five years.

[17] According to a 2017 Pew Research survey on gun ownership (Igielnik and Brown (2017)), 67% of gun owners cited protection as a major reason for getting a gun and 26% of gun owners say they regularly carry a gun outside of the home.

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three main datasets. First, a 2012 snapshot of the registry of concealed handgun permits which details individuals who received a CHP in the previous five years. Prior to 2014, Mecklenburg County, NC provided data on individuals with concealed handgun permits (CHPs). This data contains identifiable information such as full name, date-of-birth, residential address, date permit issued and if this is a new permit or a renewed permit for the universe of active permit holders. Few permits expire (10%) and even fewer permit application are denied (1%) during my study period of 2007 through 2011.

I am limited to examining CHPs since pistol purchase permits (PPP) data does not contain the pistol permit issuance date and has more limited individual information. North Carolina changed public access laws to exclude gun permits in 2014 thus limiting the time frame for which I am able to directly incorporate this data to 2007 through the end of 2011, given the data was originally requested in 2012. Figure 1 provides the frequency of new CHPs over time.[18] Two things to note: 1) the initial spike in CHPs due to the introduction of CHPs in 1996 and the large spike in CHPs after the inauguration of Barack Obama in early 2009. The inauguration of Obama generated a large spike in the demand for guns as individuals feared that the federal government would pass policies limiting gun purchases or even the creation of a national registry of new gun owners.[19] The shooting of US Representative Gabrielle Giffords in January 2011 produces a noticeable uptick in CHPs as legislation was introduced to expand background checks and additional gun restrictions.[20] These spikes become even more pronounced after the end of my CHP dataset with events such as the mass shooting at Sandy Hook Elementary in 2012.

In order to better understand how the number of CHPs compares to background checks for most firearms as well as construct my instrument in later analysis, I incorporate data from the National Instant Criminal Background Check System (NICS). This database provides the number of background checks by state and month since 1998 and disaggregates into permit and purchase background checks for handguns and long guns. I focus on handgun background checks and those associated with permits and purchases. Figure 2 provides the number of monthly CHPs and NCIS background checks for handguns in Mecklenburg County over my study period.[21] This figure has two purposes. First, this

[18] Throughout this paper, I will define the date of a CHP based on when the permit is issued, not the application date. This measure of CHP will better match both the timing of background checks as well as when an individual is able to purchase a new gun.

[19] See Depetris-Chauvin (2015) for an in-depth study of gun purchases around the inauguration of Barack Obama in 2009.

[20] Appendix Figure A1 confirms that the narrative around gun bans/restrictions and legislative actions on guns spiked around the same time as the upticks in CHPs by examining Google trends for searches that include gun bans.

[21] Since, NCIS is only available at the state level, I use a general population share to scale down NCIS counts for the state in order to indicate the total for Mecklenburg County (Charlotte, NC).

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figure shows us that background checks correspond well with the timing of CHPs and are often conducted just prior to a CHP being issued.[22] Second, I can later use this figure to scale up the number of CHPs to estimate the number of total background checks which provides an estimate of total guns. From this figure, one would expect roughly 4 background checks for every CHP and based on survey estimates of gun sellers (Brauer et al. (2013)) , these 4 background checks would correspond to about 4.4 new guns.

In order to further explore the relationship between CHP holders and other gun purchasers, I can leverage a dataset on pistol purchase permit (PPP) holders over this time period.[23] This data provides enough information to assign individuals who received PPPs between 2008 and 2012 to neighborhoods of residence and represents the main avenue for purchasing legal guns, outside of CHPs and private/gun show purchases, in Charlotte. Figure 3 provides the distribution of PPPs and CHPs across neighborhoods in Charlotte based on the main dimensions upon which I will later define neighborhoods - share of Republicans within each Census Block Group (CBG) neighborhood. Results highlight a consistent positive relationship between the number of CHPs or PPPs in a neighborhood and the concentration of Republicans. The similarity in neighborhoods for CHP and PPP holders as well as similar timing of CHPs and background checks in Figure 2 is reassuring that CHPs holders are representative of gun-buyers more broadly.

The primary dataset to link political affiliation to CHP holders and neighborhoods is the universe of registered voters based on annual snapshots from 2005-2013. This publicly available dataset provides annual snapshots of registered voters and contains a number of useful details including full name, residential address, birth year, political affiliation, race, age and gender. This dataset allows one to create a panel of registered voters including neighborhood of residence. These snapshots are typically taken right around annual elections in October and November and thus individuals who are politically active will have updated any new residence by the time of the snapshot if they were planning to vote in the upcoming election.[24] The process for updating voter registrations and voting more broadly during this time period is not onerous even though more recently (after my study period) there were a number of state legislative changes that required voter identification as well as restrictions on early voting.

The main crime dataset incorporated into my analysis is the universe of reported

[22] The two time series in this figure generate a correlation of 0.34.

[23] Pistol Purchase Permit (PPP) data was provided from the reporters in Off and Henderson (May 12, 2013) and provides individual information on anyone who obtained a PPP in the last 5 years but no info on the timing of purchase.

[24] Any concerns about updated voter registration information is partially mitigated by the fact that I attempt to merge unmatched individuals across any year of voter registration records using the uniqueness of full name and birth year. Additionally, the high match rates using my primary matching algorithm of unique full names, age and residential address limits this concern.

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crimes from the Charlotte-Mecklenburg Police Department (CMPD) which contains exact latitudes and longitudes of reported crimes, whether a crime was cleared by arrest, details about the type of crime and even details about crime victims (including names, date-of-birth, address of crime victims). This dataset of reported crimes spans 2005 through 2013 and allows the creation of detailed crime outcomes that can be gun-specific such as violent crime using a firearm as well as if specific CHP holders are later crime victims or arrested for any crime. The detailed nature of the dataset allows me to link up crime victims to registered voters even if the victims changed residences.

The final dataset I incorporate into this analysis is a database of property records obtained from Mecklenburg County which contains property transactions (e.g. sale price, sales date, loan amount) and property attributes (e.g. square footage of living area, age of home, beds, baths, etc.). I subsequently merge in transaction level data from HMDA (Home Mortgage Disclosure Act) in order to provide a measure of homebuyer income for an individual based on the most recent property transaction at a individual's given address. The merge between property records and HMDA data, which contains income for homebuyers, is discussed in the appendix with greater details provided in a separate technical appendix.[25]

Given the individual nature of my data, I begin with the creation of an annual panel dataset of registered voters linked up to CHPs based on full name, birth year and address.[26] Approximately 86% of CHP holders can be directly merged to registered voters based on name, age and address. This high match rate indicates that almost all gun owners are registered voters and have a number of demographic attributes that correlate with political participation.[27] I lose approximately 4% of the original voter data due to missing or incomplete geographical information. Since my focus is on the impacts of legal handgun ownership (which requires no felony convictions) on crimes, I am less concerned with other scholars (Uggen and Manza (2002), Hjalmarsson and Lopez (2010)) finding that criminals are less likely to be registered voters.

Table 1 provides summary statistics for this annual dataset of registered voters merged with CHP, crime and property data. This table highlights a couple interesting differences between individuals who have CHPs and those that do not. First, CHP holders are more likely to be white, Republican and older. These permit holders also tend to have been registered voters for a longer time period (likely lived in NC longer) and have slightly higher levels of individual crime victimization, but live in neighborhoods with lower levels of reported crimes. Appendix Table A1 provides some initial analysis to

[25]This technical appendix is available at https://sites.google.com/a/colorado.edu/stephen-billings/code
[26]See Appendix A1 for details on data merging
[27]Registered voters in Mecklenburg County, NC represent about 75% of the population aged 18 and above.

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formalize the differences between CHP and non-CHP holders using a cross-section of my dataset based on the January 2007 voter snapshot. This Table provides OLS analysis based on a dependent variable equal to 100 if an individual ever obtained a permit between 2007 and 2011. In Columns 1 and 2, I examine what factors explain ever getting a CHP or a new CHP with column 3 including neighborhood (Census Block Group 2000) fixed effects. Across all models, individuals that are white, Republican, older and born in NC are more likely to get a CHP.[28] In terms of neighborhood attributes highlighted in Columns 1 and 2, those neighborhoods that have lower minority concentration, less education and higher income tend to have more people obtain CHPs. Across columns, being a Republican has the largest magnitude of impacts on getting a new CHP. Interestingly, neighborhood crime levels in 2006 do not explain getting a CHP.

One additional fact to highlight is the general crime and CHP trends that exist in this data. Figure 4 provides a time series for my two main variables of interest - reported crimes and new CHPs.[29] I include reported crimes from 2005 to 2013 to highlight pre-trends as well as capture longer-term effects on reported crimes as they relate to CHPs issued during my main time period of 2007-2011. The general trend in crime from 2007 through the end of 2011 is a decline in reported crimes while there is a general increase in CHPs. It will be important to detrend this data series in later analysis by comparing across neighborhoods in the same time period. These crime trends hold for violent and property crimes as well as other crimes in aggregate.[30]

5. Individual Analysis

One of the main concerns in estimating any model that links an individual obtaining a new CHP to later crime victimizations is that crime risk is different and likely greater for individuals who obtain a CHP. The literature provides some evidence of this concern with Depew and Swensen (2019) , Tannenbaum (2020) and Costanza et al. (2013) showing positive impacts of crime on permitting activity in smaller cities in NC as well as Connecticut and New York. Therefore, one wants to generate a valid control group that will account for the demographics of individuals as well as both neighborhood and

[28]These attributes are consistent with existing research (Glaeser and Glendon (1998) , Costanza et al. (2013) , Thompson and Stidham (2010)) which highlights that being Republican, white, older and male correlate with gun ownership using survey data.

[29]I have historic monthly counts of CHPs issued but there is no identifying information for CHPs prior to 2007 in my dataset.

[30]I define violent crimes as arson, (aggravated and simple) assault, manslaughter, rape, robbery and define property crimes as auto theft, burglary, forgery/fraud , hit and run, larceny, stolen property and vandalism. These definitions expand FBI violent or property indexed crime categories to include a larger set of crimes that are often reported to local police and likely incur social costs. Later results would be quite similar with the more narrow FBI definitions.

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individual crime risk.

In order to create a control group, I implement a strict matching and difference-in-difference estimator. Specifically, I take all individuals who ever obtain a new CHP between 2007 and 2011 and match up to 10 non-CHP holders based on exactly matching all the individuals attributes given in Table 1 as well as the CBG neighborhood of residence in the month for which the CHP holder received a new a CHP. Matching is done with replacement such that a non-CHP holder may be matched to multiple CHP holders. This month provides the timing of treatment for later event study plots and difference-in-difference estimates. I further match exactly on the total number of previous crime victimizations an individual was subject to up to five years prior to the month of treatment. This final matching dimensions is the best available measure of crime risk. In order to alleviate concerns that previous crime victims alter their activity space and protective behaviors more broadly, my preferred specifications focuses only on individuals that were never previous crime victims.

In the end, I am able to match 8,420 new CHP holders to an average of 2.1 control observations.[31] I observe these matched groups of CHP and non-CHP holders for the 12 months prior and following the timing of when a CHP is issued to allow me to observe criminal victimization outcomes for a full year pre/post treatment. I begin by estimating a series of figures that provide coefficients for each month pre/post treatment for different crime categories using Equation 1.

$$y_{injt} = \alpha + \Sigma_{t^+=-12}^{12} \beta_{t^+} D_{njt^+} * CHP_i + \gamma_{njt^+} + \delta_{nt} + \epsilon_{injt} \tag{1}$$

where y_{injt} represents criminal victimization outcomes, β_{t^+} indicates the main coefficients of interest and highlights the additional effect on victimizations for CHP holders in the 12 month period prior to and after obtaining a CHP in time $t^+ = 0$. D_{njt^+} is a dummy for the time period relative to treatment (t^+) for group j in CBG neighborhood n. CHP_i is my measure of treatment and indicates if an individual ever obtained a new CHP. In order to limit comparisons between treatment and matched control observations in the same time period relative to treatment, I include fixed effect γ_{njt^+}. Finally, to control for general neighborhood time trends in victimization, I include a fixed effect for neighborhood by month of the observation (δ_{nt}). I cluster standard errors at the level of the matched group as well as the census block group by month.

Result of this estimation for the full sample of matched CHP and non-CHP individuals is provided in the top panels of Figure 5. This figure highlights no presence of

[31] I lose approximately 158 CHP holders that do not have any valid control matches.

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pre-trends in crime victimization as well as elevated crime victimization for property crimes, but not for violent crimes. Effects for property crimes are strongest in the months immediately following the new CHP. When I narrow the sample down to only individuals without prior victimizations in panels C and D, results show the mechanical fact that pre-treatment coefficients are zero by definition and that post-treatment outcomes look quite similar as the full sample. Finally, panels E and F provide some gun-specific crime victimizations with results for stolen firearms generating large positive coefficients and violent crimes using a gun generate effects close to zero. Given that these types of crime victimization are less frequent, I collapse time periods into quarters.

In order to provide a magnitude for the effects of obtaining a new CHP on crime victimization, I estimate a two period difference-in-difference estimator given by Equation 2 where β_1 is the main coefficient of interest and $D_{nj(t^+ \geqslant 0)}$ is equal to one for the month and subsequent 12 months after a CHP is issued to group j.

$$y_{injt} = \alpha + \beta_1 D_{nj(t^+ \geqslant 0)} * CHP_i + \beta_2 CHP_i + \gamma_{njt^+} + \delta_{nt} + \epsilon_{injt} \tag{2}$$

Results from Equation 2 are provided for a range of criminal victimizations in Table 2. Panels A and B highlight the relationship between getting a CHP and crime victimization for all as well as just individuals that were never crime victims. Focusing on my preferred estimates in Panel B, getting a new CHP is associated with a 68% increase in being a crime victim. In terms of gun specific crimes, a new gun owner is 228% more likely to have a gun stolen. Results for just individuals with prior victimizations are given in Panel C and offer a different population of potential gun owners and victims. Results are quite noisy for this sample, but coefficients are positive and somewhat consistent with other results in this table. All together, these results suggest getting a CHP increases property crime victimization with additional victimization for gun-specific crimes.

In order to highlight the potential of CHP holders being involved in transferring guns to criminals or using guns for criminal activities, I provide results for estimating Equation 2 by simply replacing the outcome to being arrested and restricting subsamples based on prior arrests. The results in Table 3 indicate that individuals who obtain a CHP are less likely to be arrested, even though results for a number of arrest types are underpowered.[32] The fact that CHP holders are less likely to be arrested is suggestive of a general narrative that individuals who legally obtain a gun may be more law abiding individuals and this only increases after getting a CHP. This result does not dispel the presence of straw purchasers since individuals who legally obtain guns to sell may take

[32] Appendix Figure A2 confirms no pre-trends in violent or property crime arrests.

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effort to avoid criminal detection and thus obtain fewer arrests. Overall, these results do suggest that any concern about selection on unobservables for obtaining a CHP indicate that they would be along the dimensions of law abiding behavior. These results, if anything , would correlate with behavior that leads to less crime victimization as CHP holders likely avoid situations involving criminal activities.[33]

Since the largest impact of obtaining a CHP is on stolen firearm victimization, there is a large scope for impacts beyond the individual CHP holder. I therefore turn to neighborhood analysis of the relationship between guns and crime. Individual results are informative in highlighting crime outcomes that vary between individual and neighborhood scales and thus speak to the underlying mechanisms that may translate more guns into crime.

6. Neighborhood Crime

Establishing a causal link between CHPs and neighborhood crime has a number of challenges including concerns of reversal causality between recent crime and getting a permit as well as omitted variable bias due to residential sorting and neighborhood trends. In order to overcome these empirical issues, I adopt an instrumental variable methodology which is similiar in spirit to a substantial literature that uses differences in initial conditions and exogenous shocks as instruments.[34] Specifically, I use the initial share of Republicans in a neighborhood in 2005 and subsequent shocks to the demand for CHPs due to national events (e.g. election of Obama, Rep. Giffords shooting) and the subsequent legislative discussions about limiting gun access.

Therefore, variation in my model is due to exogenous events that increase demand for guns and the dosage of these shocks are based on a neighborhood's initial endowment of likely gun owners - Republicans.[35] There are some concerns with this type of identification that have been discussed in recent papers by Goldsmith-Pinkham et al. (2018) and Borusyak et al. (2018). I will follow some of their suggested specification tests as well as add some additional tests to demonstrate the validity of this instrument. Before highlighting the validity of the shift share instrument, let me formally define it.

In this neighborhood analysis, all individual data will be aggregated into a neigh-

[33] A simple correlation between being a crime victim and arrested for a crime indicate a positive relationship of 0.08 for all crimes and 0.09 for violent crimes.

[34] Some highly cited papers from this literature include Currie and Gruber (1996) , David et al. (2013) , Nakamura and Steinsson (2014)

[35] I chose Republicans versus other attributes since this individual attribute was the strongest predictor of getting a CHP. I also created a composite initial share based on the share of individuals that are older, Republican, white , born in NC and this provided quite similar results which I discuss later. These are all attributes associated with getting a CHP based on Appendix Table A1.

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borhood monthly balanced panel from 2007 through the end of 2011 where neighborhood will typically be defined as Census Block Group (CBG) 2000. Defining neighborhoods is always challenging given that neighborhoods are conceptualized in so many different ways and their size may also depend on the nature of the outcome under examination. The initial choice of CBG neighborhoods is justified for looking at crime outcomes because the CBG size encompasses the typical distance upon which a criminal travels from their home. Specifically, Townsley and Sidebottom (2010) state "Distance-to-crime research is summarized by two findings, most crime trips are short and offenders do not travel far to commit crime." The average distance traveled is similar to the average diameter of a CBG 2000 neighborhood in my study area.[36]

The initial OLS estimation equation is given by Equation 3 with n indicating one of 373 CBG neighborhoods and t indicating month. To address potential delays from when an individual is issued a new CHP and when they obtain a gun, I define the main variable of interest over a two month window where $CHP_{n,[t,t-1]}$ indicates the total number of new neighborhood CHPs in month t=-1 through t=0. Correspondingly, $X_{n,t-1}$ indicates previous year neighborhood measures of annual voter attributes such as share of voters that are black, Hispanic, age categories (<30, 30-50, 50+), longer-term voter (registered at least 5 years ago) shares and homebuyer average income. The inclusion of neighborhood fixed effects (γ_n) accounts for any general population or voter differences across CBGs and limits identification to changes in crimes over time. Furthermore, δ_t provides fixed effects for each month and thus controls for monthly crime levels in the study area of Mecklenburg County, NC. $y_{n,[t,t+k]}$ provides crime outcomes in time t with a time window upon which to observe neighborhood crime that will depend on chosen values of k, which I discuss in the next section.

$$Y_{n,[t,t+k]} = \alpha + \beta CHP_{n,[t,t-1]} + X_{n,t-1} + \delta_t + \gamma_n + \epsilon_{nt} \tag{3}$$

For instrumental variable analysis, I implement a shift share instrument that is based on two main components. The first component is the share of all Republicans in a neighborhood and is defined as $ShareRep_{n,t=2005} = \frac{Rep_{n,t=2005}}{\Sigma_l Rep_{n=l,t=2005}}$. The fact that I incorporate 373 neighborhoods into this analysis limits concerns about overrepresentation by a few large cross-sectional units raised by Goldsmith-Pinkham et al. (2018). Appendix Figure A3 provides a histrogram of share Republican and shows that many

[36] A back on the envelope calculation weighted by the population of registered voters finds the average CBG to be 2.4 square miles or a diameter of 1.5 miles which corresponds well with the distance to crime literature. I also provide results for larger neighborhood definitions based on tracts which average slightly larger diameters than this literature at about 2.5 miles.

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neighborhoods contain a nontrivial share of Republicans and that the most Republican neighborhood represents 2% of all Republicans in my study area. The second component of the shift share instrument is the flow of new CHPs over time, which over the time period is heavily influenced by national shocks to the demand for guns given by the inauguration of Barack Obama as well as shootings events and legislative discussions that led to increased awareness and concern about future loss of gun rights. In order to remove any influence of local demand shocks in Charlotte, NC, I measure national shocks based on total handgun background checks (NICS) for all states except North Carolina in time period t and $t-1$ ($\mathrm{NonNCNICS}_{[t,t-1]}$). Therefore, local shocks in Charlotte or even North Carolina will have no influence on this instrument. Combining these two components is given by Equation 4 where β_1 is the main coefficient of interest. Given the specification of my instrument in terms of one dimension (share Republican), inference concerns highlighted by Adao et al. (2019) are addressed by clustering standard errors along the dimension of initial Republican shares. Specifically, I include standard errors clustered by neighborhood and also at the month by initial neighborhood share of Republicans (binning shares into 20 equally sized groups) level. This generates 1,200 (20*60) clusters given the 60 months of data.

$$
\begin{aligned}
Z_{n,[t,t-1]} &= \mathrm{ShareRep}_{n,t=2005} * \mathrm{NonNCNICS}_{[t,t-1]} \\
\mathrm{CHP}_{n,[t,t-1]} &= \alpha + \beta_0 Z_{n,[t,t-1]} + \delta_t + \gamma_n + \epsilon_{nt} \\
Y_{n[t,t+k]} &= \alpha + \beta_1 \widehat{\mathrm{CHP}_{n,[t,t-1]}} + \delta_t + \gamma_n + \epsilon_{nt}
\end{aligned}
\tag{4}
$$

The issuing of CHPs and the timing of NICS background checks appear similar as shown in Figure 2 and later specification curve analysis will highlight limited variation in my main result if I allow background checks to vary by a month or two from actual CHPs. Given this is an instrumental variable approach, I need to discuss its relevance and provide some evidence of exogeneity. Highlighting the relevance of my instrument is seen in Figure 6 where I compare CHP frequency over time for neighborhoods with above median share Republicans (High Repub neighborhoods) relative to below median share Republicans (Low Repub neighborhoods). The relevance of this bisection is clear from the substantially larger share of CHPs in Republican neighborhoods and this holds well through the large spike in early 2009 from the Obama Inauguration as well as for later shocks. Beyond this graphical evidence, later regression results will highlight the large F-statistic generated from the first stage of Equation 4.

Providing evidence on exogeniety is quite a bit more complicated, but the main concerns about this type of instrument is not the exogeniety of the national shocks, but

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rather how I define the dosage of these shocks based on share Republican. The main issue being differential trends for Republican versus non-Republican neighborhoods especially since there is a clear upward trend in CHPs over the study period. I implement a couple tests of this concern. First, as discussed by Goldsmith-Pinkham et al. (2018) , one would like to see a figure that shows parallel pre-trends when implementing a shift share instrument. One would not expect this to hold completely in my context since new CHPs were issued before my study time period, but as shown earlier, the number of CHPs are quite a bit lower prior to the large uptick at the end of 2008/early 2009. In order to formally implement a test for pre-trends, I estimate a time series specification where I interact my continuous share variable ($ShareRep_{n,t=2005}$) with a dummy for each month of my study period. Specifically, I estimate Equation 5 for property and violent crime outcomes where I include neighborhood as well as month fixed effects.

$$y_{nt} = \alpha + \beta_1 \Sigma_{t=Jan07}^{Dec11} D_t * ShareRep_{n,t=2005} + \delta_t + \gamma_n + \epsilon_{nt} \tag{5}$$

Results of estimating Equation 5 along with a histogram of new CHPs issued is given in Figure 7. This figure highlights no clear pre-trends for property or violent crimes in the months preceding the large uptick in CHPs around the election and inauguration of Barack Obama and foreshadows later results of elevated violent crimes following the Obama bump in CHPs.

The use of an instrument constructed using the total stock of CHPs in other Charlotte neighborhoods for each month (instead of Non-NC NICS counts) is more similar in spirit to an international trade based shift share instruments but I do not use this version of the instrument to avoid any concerns regarding correlated shocks across neighborhoods that could be a result of local economic conditions or defining the neighborhoods incorrectly. This concern is often referenced as the spatial independence assumption needed in shift share instruments (Goldsmith-Pinkham et al. (2018)). I later show that results are quite similar when defining the instrument this way.

Additionally, Goldsmith-Pinkham et al. (2018) suggests testing the role of control variables in impacting results. Appendix Figure A4 provides some results for mean total crime differences between low and high Republican neighborhoods with the inclusion of different combinations of annual voter attributes. This figure highlights that only race tends to shift the differences between these two types of neighborhoods, but this shift does not change over time. Therefore, neighborhood fixed effects will address any initial differences in crime and residential racial composition before the uptick in CHPs.

One additional test that can further assuage any concerns about differential trends

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in low versus high Republican neighborhoods is to examine home sales prices and mortgage stated income for new homebuyers across these two types of neighborhoods. If these types of neighborhoods were converging over time then the shifting crime levels that correlate with CHPs could simply be capturing neighborhood trends that differ based on share Republican. Appendix Figures A5 and A6 provides graphical evidence that low and high Republican neighborhoods appear to trend in similar ways or even slightly diverge in later years of the study period. This divergence in income or gentrification of Republican neighborhoods would, if anything, push any relationship between CHPs and crime to find that more CHPs leads to lower crime.

Since this instrument and dataset does limit my ability to test results using larger geographical units than census tracts, one may be concerned about neighborhood spillover effects muting my results. For example, if there is an additional CHP in a given neighborhood that results in a stolen gun which is used to commit crimes in another neighborhood, effects could be downward biased. This concern rests on two assumptions. First, criminals have to be active across multiple different neighborhoods and thus exhibit a longer journey-to-crime (JTC) then is suggested by the literature. Second, even if the JTC is sufficiently long then a necessary condition for results to be biased is that a stolen gun obtained in one neighborhood (presumably a higher CHP neighborhood) must be used to commit a crime in a low CHP neighborhood.[37] Luckily, I can at least test these two conditions for a subset of reported crimes that led to arrest.

Appendix Figure A7 shows that distance between an arrested individual's home and the corresponding reported crime that led to arrest for all crimes as well as violent crimes and just stolen gun reported crimes. I limit analysis to up to 10 miles apart since less than 5% of observations register a longer journey to crime. The median distance from an arrestee's home address and the reported crime location that led to arrest is 1.9 miles for all crimes, 0.38 miles for violent crimes and 1.25 miles for stolen firearms.[38] Additional analysis examines if individuals who commit crimes in other neighborhoods do so in neighborhoods with different levels of CHPs. A simple test is calculating the percent of arrestee and reported crime pairs for which the reported crime neighborhood is above/below median numbers of CHPs and the arrestee home address is the opposite level of CHPs. This calculation provides a sense of potential bias from neighborhood spillovers. I provide this calculation for a range of crime types in Appendix Table A2.

[37] In essence, if criminals are stealing guns and using them to commit crimes in other neighborhoods from where they stole the gun as long as these two neighborhoods are similar, estimates will exhibit limited bias.

[38] Given these distances are calculated based on arrested individuals, one could imagine non-cleared reported crimes could be be shorter or longer distances. Friends and neighborhoods could be less cooperative with the police for short JTC thus undersampling short distances, but criminals that travel further may make it more difficult for the police to find the offender.

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This table provides evidence of some neighborhood spillovers but that only between 20% and 30% of incidents should led themselves to providing any downward bias to my main estimates. Of course, potential spillover effects decrease for models with tracts and as I later show these larger geographies present quite similar results.

7. Neighborhood Results

In order to implement this instrumental variable analysis, I need to first determine the temporal scale for measuring the impacts of CHPs on total crime (i.e. what value of k to use in Equation 4). Even though I examined the temporal scale of crime vicimization in earlier results, effects of neighborhood crime may evolve quite differently since neighborood crimes encompass all criminal activity as it evolves after the introduction of the CHP. A priori, it is not obvious how long the effects of a CHP and corresponding gun purchases may last on neighborhood crime. An initial gun purchase may immediately lead to the transfer of guns to the illegal market.[39] Subsequent months may also provide additional effects as legal guns are used for criminal activity as well as continue to be available to gun owners. Different crime outcomes may also have quite different temporal impacts. In order to provide some guidance on the criminal outcome window ($t = 0$ through $t = k$), I estimate Equation 4 separately for different crime outcomes and unique values of k as given by the x-axis in Figure 8.

This figure provides two insights. First, for most crime categories, one sees a larger immediate impact suggesting the importance of examining short outcomes windows.[40] For some crime categories, one also sees impacts at longer time windows (k>2) suggesting some non-immediate impacts. Time periods further out are often similar or weaker in impacts and also more likely to capture future spikes in CHPs as well as changing neighborhood dynamics. Together these figures suggest examining impacts that are immediate (t=0 through t=2) and medium term (t=3 through t=5) and subsequent results will highlight both as well as a composite impact of 6 months (t=0 through t=5).[41]

Table 4 provides results for a number of relevant measures of crime given by row headings. Table 4 provides three panels of results. The OLS model with fixed effects is given in Panel A and generates insignificant and small effects on total crime. The

[39] Cook and Goss (2014) discuss how legal guns are often sold on the illegal markets and that some legal gun owners purchase additional guns simply to sell for profit.

[40] The presence of large immediate positive impacts of CHP may also capture the presence of pistol purchase permits and other gun sales that are processed quicker than CHPs and with large upticks in background checks as shown in Figure 2 this could mechanically generate crime effects as CHPs are being issued.

[41] Given that there are not clear cutoffs in these figures, I also provides results for other outcome windows in Table A5.

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second panel provides estimates with the shift share instrument based on NICS handgun background checks for non-NC states and neighborhood share Republican in 2005. The estimates in panel B provide results for three months of crime outcomes and indicate that one additional CHP (about the average of 2.5 months of neighborhood CHPs) generates an increase of 4.2 total crimes. Interpretation is really more accurately scaled by NICS background checks. Each CHP corresponds to about 4 NICS background checks which according to Brauer et al. (2013) generates 4.4 handguns. Therefore, on average each additional CHP correlates with approximately 4.4 more guns which corresponds to 7.0% (4.2/59.1) more total crime in the first three months. Subsequent crime in the next three months indicate effects sizes that are about half the size and less precise. Focusing on the preferred 6 month window, I find an increase of 5.3% (6.3/117.6) total crime.

The downward bias in OLS estimates in panel A is surprising since a priori one might expect any bias would be a story of higher neighborhood crime risk leading an individual to obtain a permit and this would generate a positive bias due to elevated pre-CHP crime risk. In order to explore this paradox, Appendix Section A.2 provides a more thorough analysis of the relationship between neighborhood crime, permitting and moving and does provide some explanation for why one may see a downward bias in OLS results. In essence, individuals experiencing high neighborhood crime risk are more likely to move away and get CHPs. This correlation would generate a negative relationship between existing crime risk and getting a CHP in that neighborhood as likely gun owners move away. Another possible explanation for smaller OLS results is mean reversion as high neighborhood crime in one year may lead to more individuals getting CHPs and crime simply reverts back to its longer-run average in the following year. It is difficult to confirm this mechanism since mean reversion may not occur if there is a strong impact from CHPs on crime. One piece of evidence that limits a strong role for mean reversion is that there appears to be limited evidence of upticks in recent neighborhood crime leading to increases in CHPs.[42]

Since total crime encompasses many incidents that are unlikely to be related to guns, the remaining columns of Table 4 further refine crime outcomes to those that are likely to be related to guns. Each additional CHP generates a 4.8% (0.97/20) increase in violent crimes and an imprecise 4.1% (3.0/74.4) increase in property crimes for 6 month crime outcomes. Temporal effects are still strongest for the initial three months of crime outcomes. Focusing on Panel D, columns 4 through 6 provide some gun relevant outcomes. Crimes that produce a serious injury victim (which could be due to conflict escalation or the changing composition of crime as guns are introduced to the neighborhood) generate

[42] Appendix Table A3 shows limited evidence of any relationship between lagged neighborhood crime and an individual getting a CHP.

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an effect size of 0.25 and represents about a 16% increase for each additional CHP. Crimes where a gun was stolen in column 5 are the most direct measure of the potential impact of legal guns as criminals are likely to steal guns through burglaries, theft from auto or robberies (if a person is carrying a gun).[43] Consistent with individual results, this outcome also generates large effects with each additional CHP increasing crimes with a gun stolen by 20%.[44] Column 6 provides a direct measure of using guns illegally and finds that crimes using a gun increases by 32% for each additional CHP and supports the narrative of additional legal guns entering the illegal market. Unlike more aggregate crime categories, these three crime outcomes generate effects in both the shorter and longer time window of examining crime outcomes.

To formally test if crimes are shifting in composition, to crimes that involve guns, Table 5 divides gun-related crimes by corresponding violent or property crimes. These new dependent variables highlight if the increases in gun-related crimes in Table 4 are beyond the general increases in violent or property crimes. Columns 1 of Table 5 shows that serious injury victim crimes are imprecisely increasing as a share of violent crimes indicating that greater injuries are primarily of function of more violent criminality in the neighborhood than escalation. Column 2 highlights that stolen guns are also not increasing beyond property crime increases. This result could indicate that theft may increase as the stock of higher valued items like guns increase, but criminals are not targeting gun owners specifically in their crimes, but rather yielding guns as part of their crime. Column 3 indicates a compositional shift to violent crimes using guns with each additional CHP generating a 3.5 p.p increase in the share of violent crimes using a gun (a 20% increase). In essence, results for gun-related crimes are a mix of compositional shifts toward crimes that involves guns along with some overall increases in violent and property crimes. Column 4 provides evidence that non-gun weapon crimes are unchanged in this environment consistent with the shift towards violent crimes using guns.

A common concern in any study of neighborhood crime is the role of policing behavior or changes in the composition of crimes reported by victims.[45] Policing behavior may shift to gun-related crimes as officers are concerned about the increased number of new legal guns shifting to illegal activity. This shift in policing has two main effects,

[43] Based on CMPD data, 14,092 guns were reported stolen over my study time period in Charlotte-Mecklenburg.

[44] Unlike some other measures of crime, under-reporting is less likely when a gun is stolen because a legal owner wants to make sure future crimes committed with a stolen gun are not attributed to the victim of this theft. Additionally, guns are high valued items and thus more likely to be claimed on homeowners or renters insurance.

[45] In equilibrium, one expects that policing readjusts to deal with increased crime levels and that the main driver of changes in crime reporting is shifts in the types of actual crimes.

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increasing the number of reported gun-related crimes as well as deterring gun-related crimes as more officers are nearby. The joint effect of targeted policing on gun-related reported crimes is ambiguous but should lead to a short-term positive effect (and eventually no effect in equilibrium) on crime clearance if officers are more focused on gun-related crimes.[46] Results in columns 5 and 6 of Table 5 suggests neither effects on total crime clearance nor gun-related crime clearance in the short-term, which does limit a large police response from making more officers available to investigate and clear the increased gun-related crimes.

Given the strong effects of CHPs on stolen firearms from individual as well as neighborhood analysis, Table 6 provides additional results for different categories of gun crimes including crimes with a gun stolen from a car, home and other locations, if a gun theft involved multiple guns, total property stolen as well as individuals killed by a gun. I find limited effects of CHPs on guns stolen from a home or car. These results do highlight that guns stolen from other locations (which is primarily from a person or commercial locations) do appear to be related to the initial stock of legal guns. This result is consistent with the immediate timing of new CHPs on neighborhood crime more broadly. A person who obtains a new CHP may initially carry their new gun on them given they have just obtained this new legal status. This new gun effect may lead to quite contemporary timing for gun theft and other crimes. Additionally, theft from commercial locations would be consistent with a large stock of guns being made available for the large uptick in gun demand which drives my variation in new CHPs. These types of theft would occur in the same time period as new CHPs and gun purchases. No effects for multiple gun thefts does provide evidence that individuals are not stockpiling guns as part of spikes in guns demand or one would expect more instances of multiple guns stolen at once. No effects on total property value stolen highlights limited overall increases in theft in terms of value. One of the most striking result in this table is that deaths using a gun increase by 47% , but this outcome is less precise and is based on 214 gun-crime deaths over this study period.

Another way to examine the impacts of CHPs is on other crime categories which are given in Appendix Table A4. Some highlights include CHPs increasing reported crimes classified as drugs, auto theft, fraud, robbery and weapons. Surprisingly, burglaries are not impacted by CHPs. The lack of impact on burglaries supports that criminals are not increasing the number of burglaries to obtain guns, but rather a burglary is more likely to yield a gun. Effects for robberies and weapons crimes are consistent with higher illegal gun activity highlighted earlier (e.g. gun theft from others and crimes using guns).

[46] As criminals re-optimize their criminal decisions after experiencing increased police focus on gun-related crimes, this will shift some criminals to other crimes or change the composition of criminals willing to commit gun-related crimes.

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Auto theft, drugs and fraud have less direct links to guns, but are still consistent with guns being introduced into the illegal market where they can lead to a variety of criminal actions such as gang related activity, drug trafficking or illegal weapon sales.

Appendix Table A5 provides some alternative models. Specifically, Panel A provides results for a larger value of k or a crime window that expands out to one year after new CHPs. These results are quite similar and generate percent impacts that follow the main results. Panel B demonstrates that results are robust to a more comprehensive version of my instrument using a composite measures of voter attributes that predict obtaining a CHP to calculate initial shares. Panel C presents a version of my instrument that limits any influence from neighboring states by excluding South Carolina, Virginia, Tennessee and Georgia. Panel D highlights a model that only uses the Obama spike by limiting analysis from June 2008 to June 2009 and using the 3 month crime outcome window. These results follow my main results and alleviate concerns that general fluctuations over my study period that may relate to broader crime trends are influencing my instrument. In order to further test the sensitivity of my instrument, control variables and fixed effects, Appendix Figure A8 provides a number of different models for total crimes that incorporate alternative version of my shift share instrument based on various measures of NICS background checks as well as a model that directly uses the stock of CHPs in other neighborhoods. This specification figure provides some reassurance that the choice of instrument as well as that the incorporation of any annual control variables from the voter registration and HMDA records do not generate large differences in the magnitude of effects. Appendix Table A6 provides my main analysis using a tract definition of neighborhood. Results are slightly smaller in effect sizes to CBG neighborhoods with some increased precision for a few outcomes.

8. Mechanisms and Social Costs

As discussed earlier, there are a number of mechanisms that may generate a positive relationship between new CHPs and crime including interstate trafficking and large scale straw purchasing, informal transfer of guns to criminals based on social connections, stolen guns and even licensed gun dealers acting outside of the law. The nature of my results does provide some limits to interstate trafficking and large scale straw purchasing as the immediate effects of CHPs on neighborhood crime is too soon to be a result of larger-scale distribution to criminals. Nonetheless, some evidence to test for guns flowing in/out of my study area would be helpful to support or dispute this mechanism. Specifically, I obtained additional data from the gun tracing database (thetrace.org) and calculated the number of guns originating in Charlotte that were later traced to crimes

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outside of Charlotte from 2010 to 2012 and found less than 3% of guns left the county as well as calculated that only 3% of guns recovered in Charlotte were from outside of the city.[47] This data does not support large flows of guns between Charlotte and other cities being the main driver of gun related upticks in crime.

The role of local straw purchases may still be important as people buy legal guns and sell/transfer to other people. There is really no way to highlight if legal gun buyers are providing guns through informal networks, especially if these transfers involve small quantities of guns. One could even imagine that straw purchasers may be less likely to get arrested in order to continue their business of selling legally obtained guns. Of course, the sale of illegal guns does generate increased risk of weapons arrest. Earlier results (Table 3) does suggest that gun buyers are largely law-abiding since individuals are less likely to be arrested for property crimes after obtaining a CHP, but we see no change in arrest risk for other crime categories. This result does provide limits to the scale upon which CHPs holders are doing illegal activities such as transferring guns to friends or selling guns to people otherwise one would see an increase in arrests after obtaining a CHP (especially for weapons arrests).

The final and most consistent mechanism highlighted in my analysis is the role of stolen guns. Results at both the individual as well as neighborhood level provide strong evidence that increases in CHPs coincide with large increases in stolen guns. In terms of individuals, the large increase in stolen guns largely explains crime victimization more broadly and in general CHP holders are only victims of property crimes. Moving to the neighborhood crime results confirms that stolen guns increased immediately following increases in CHPs and that there is also an increase in violent crimes as well as an increase in the share of violent crime using guns. This result is consistent with a mechanism of stolen guns being transferred to criminals that are using guns to commit violent crimes.

To put my results in terms of actual social costs, I estimate the impacts of CHPs on neighborhood crime for specific crime categories and combine these estimates with the costs of crime from McCollister et al. (2010) to generate a estimated impact of the neighborhood social costs of the additional crime created by an expansion in CHPs.[48] Under a number of assumptions such as linear marginal effects and that the time window for outcomes as well as definition of neighborhoods capture the full impacts of CHPs,

[47] These percentages are based on 361 guns recovered and 363 guns reported stolen in Charlotte. The percentages are quite low in Charlotte relative to other locations. For example, this dataset finds substantially larger (37%) percentages of guns originating from other locations that are eventually recovered in Chicago.

[48] Specifically, I estimate my main neighborhood model (Equation 4) for the number of reported crimes in each crime category as a separate outcome and report those estimate in column 3 of Table 7

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one could estimate the aggregate impacts of new CHPs on crime and compute a social cost. As shown in Table 7, each additional CHP would predict 13.2 additional annual crimes (twice the 6 month estimated coefficient) in a neighborhood for a social cost of $ 191,301 per neighborhood which equates to about $ 76.55 per person.[49] Since gun owners are more likely to be property crime victims, property crime costs which equate to $35.79 per person likely encompass some private costs, but may also incur social costs as gun availability may increase neighborhood criminality (which include more property crimes in general). The remaining $40.76 in violent crime costs per person are often born by other residents from earlier analysis and thus is a primarily a external cost that is not incorporated into gun ownership decisions.

Directly comparing these cost estimates to existing literature is not easy, but Cook and Ludwig (2006) find marginal social costs of between $100 and $1800 for an additional gun. Cook et al. (2000) put gun assaults at $80 Billion and $20 Billion for accidental shootings/suicides. That would translate to about $285 per person for gun-related crimes ($80B/280M). The best comparison to my results would be if one were to eliminate all new CHPs over the study period and calculate how much social costs decreased in Table 7. Eliminating all new CHPs in my dataset would be equivalent to removing 24.3 CHPs from each neighborhood which would reduce social costs by about $1,863 ($76.55 * 24.3) per capita which is not that far out of line with the existing estimates mentioned above. My estimates may be larger because I account for a larger number of crime categories than existing estimates, but I may also underestimate the social costs of a gun that is used for crimes outside my definition of neighborhoods.

9. Conclusions

Together, results highlight that gun owners are 68% more likely to be crime victims and victimization often involves having a gun stolen. There is no evidence that legal gun owners are committing crimes themselves and in anything, once a CHP is issued the recipient appears to be more law abiding. Neighborhood analysis highlights causal estimates of CHPs on crime as well as accounts for any spillover effects beyond the individual who obtains the CHP originally. The neighborhood analysis confirms the individual results that more guns are stolen as well as some additional insight into increases in violent crimes of about 4.8% for the neighborhood and these violent crimes are more likely to involve a gun (20% increase). Additional analysis using a sample of gun

[49] The fact that one additional CHP generates 13 additional crimes reflects the fact that on average each CHP corresponds with multiple background checks and gun purchases and that one gun that moves to the illegal market may be used for dozens of crimes and transferred across multiple criminals.

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tracing data over this time period as well as further explorations of stolen gun crimes and other crime categories highlight a narrative of new CHP holders being a victim of stolen guns that are then used locally in a range of criminal activities.

Context and local gun restrictions may help explain while I find a large role for stolen guns, while some other scholars (e.g. Cook et al. (2015b) , Collins et al. (2017)) stress a large role for straw purchase and transfer of guns from illegal dealers. For example, a place like Charlotte with moderate restrictions on obtaining legal guns is quite ripe for having stolen guns as a large supplier of illegal guns. Whereas a place like Chicago, which has high demand for guns in general and some of the strictest gun laws in the US, makes obtaining guns via theft more costly thus generating large incentives to bring in guns from outside the city and state. A place like Chicago also provides a larger potential return to more established arm dealers within a large city to purchase illegal guns. This difference in context is important in thinking more broadly about gun policy since it suggests that different gun restrictions lend themselves to different mechanisms through which guns can be used by criminals.

My results suggest that policies that target gun security and storage may have substantial benefits beyond limiting the theft of guns including the prevention of future neighborhood violent crimes. Policies such as Child Access Prevention laws legally mandate safe and secure storage for homes with children and have been shown to decrease child access to guns and decrease injuries (Anderson and Sabia (2018) , DeSimone et al. (2013) , Webster et al. (2004)). Safe storage laws are effective in limiting accidental deaths and suicides (Fleegler et al. (2013), Grossman et al. (2005)). For both of these commonly used policies to promote the secure storage of guns, my results indicate that the benefits of these policies may be understated as they also prevent gun theft more broadly and associated spillovers to nearby violent crime. In the near future, one could easily see the adoption of biometric recognition devices that limit gun use to only registered owners (Cook and Goss (2014)). It is unlikely that federal gun policy will ever meaningfully limit gun ownership more broadly, but stronger policies that limit gun theft may have substantial benefits on local crime and thus something that may appeal to state and county policymakers and policing agencies.

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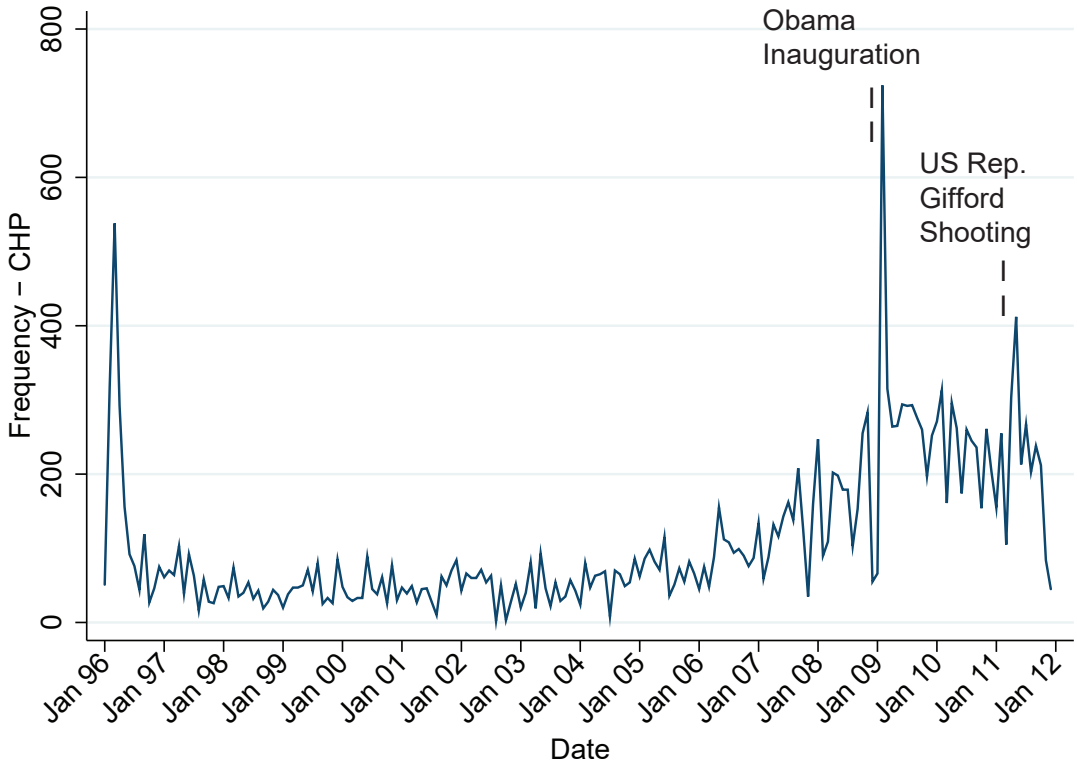
Figure 1: New Concealed Handgun Permit Holders



This figure provides the frequency of new CHPs issued from 1996-2011. Month of CHP is based on the date the permit was issued to an individual.

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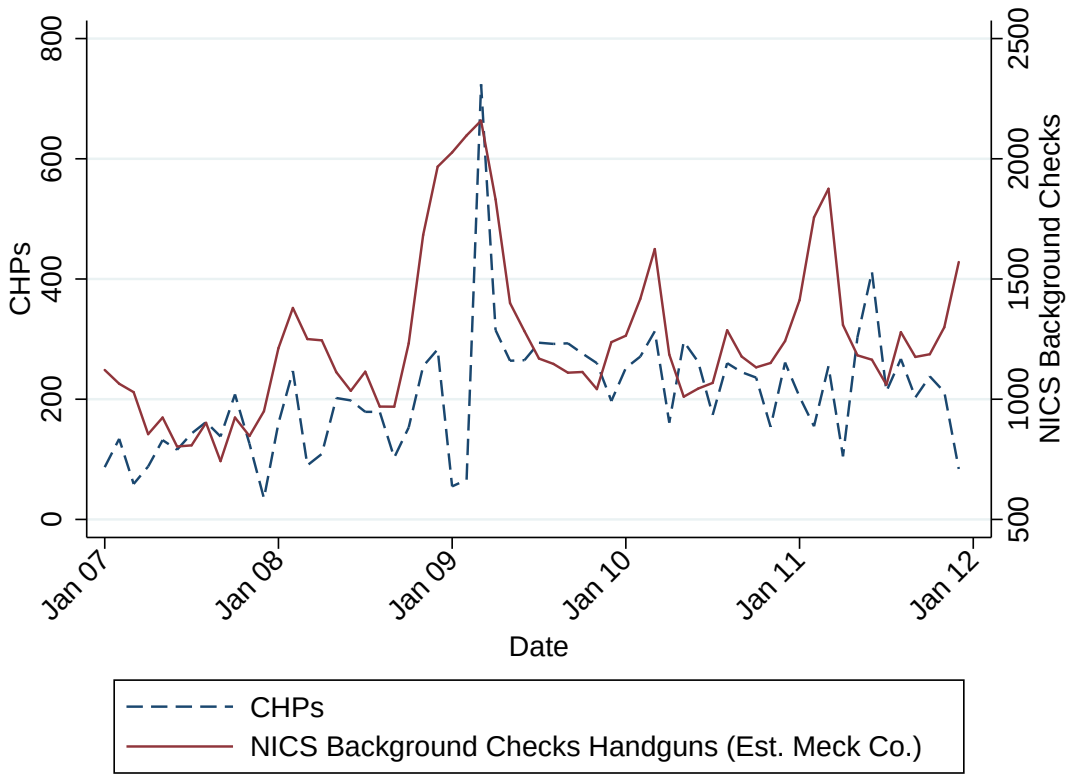
Figure 2: New Concealed Handgun Permit Holders vs. NICS Handgun Background Checks



This figure provides the frequency of new CHPs issued versus estimated NICS handgun background checks for gun permits for Mecklenburg County, NC. NICS background checks for gun purchases are provided monthly at the state level, so I estimate for Mecklenburg County by taking the share of NC population in Mecklenburg County and multiplying it with the total NICS handgun background checks for the state. These two time series have a correlation of 0.34.

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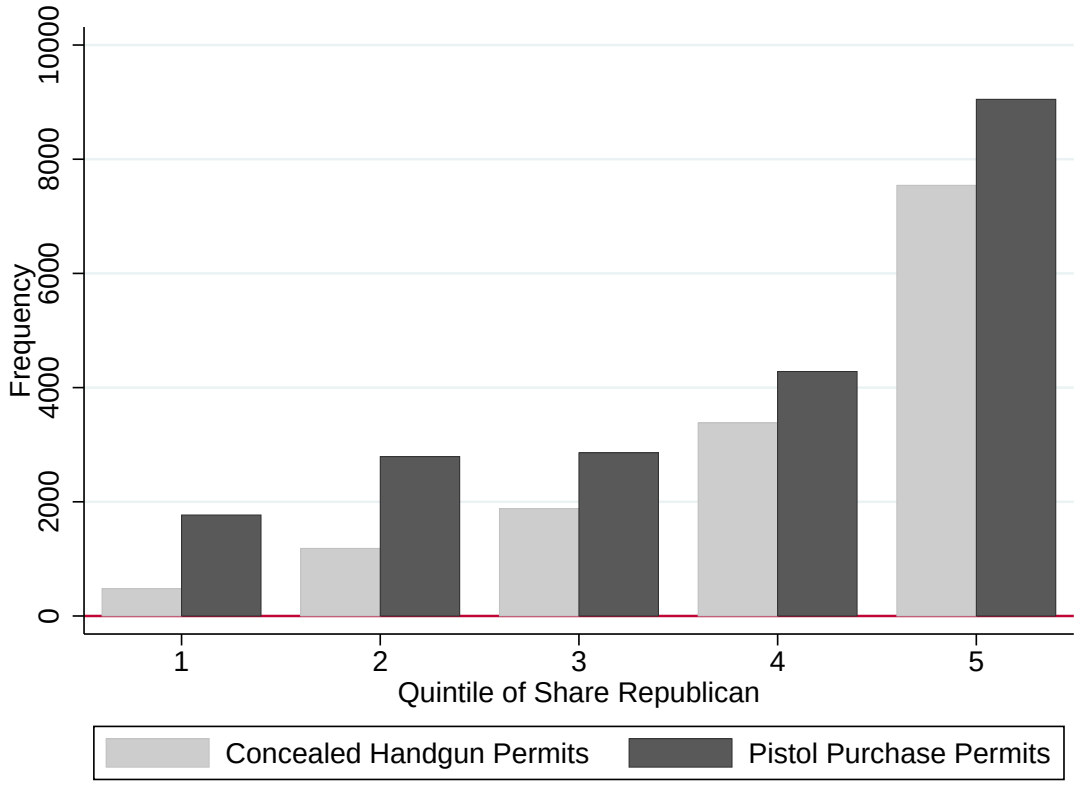
Figure 3: Concealed Handgun Permit Holders vs. Pistol Purchase Permit Holders



This figure provides the distribution of CHPs issued 2007-2011 relative to PPPs issued from 2008-2012. Since I cannot distinguish if an individual has previously obtained a PPP, I include all CHP holders regardless if they are new or renewing. There are a total of 14,468 CHPs issued and approximately 20,749 PPPs issued over the five year time period. This histogram is split along quintiles based on a CBG's share of Republicans (which is the share used in my shift share instrument). This figure highlights that the population of PPP holders live in similar neighborhoods defined by political affiliation as CHP holders.

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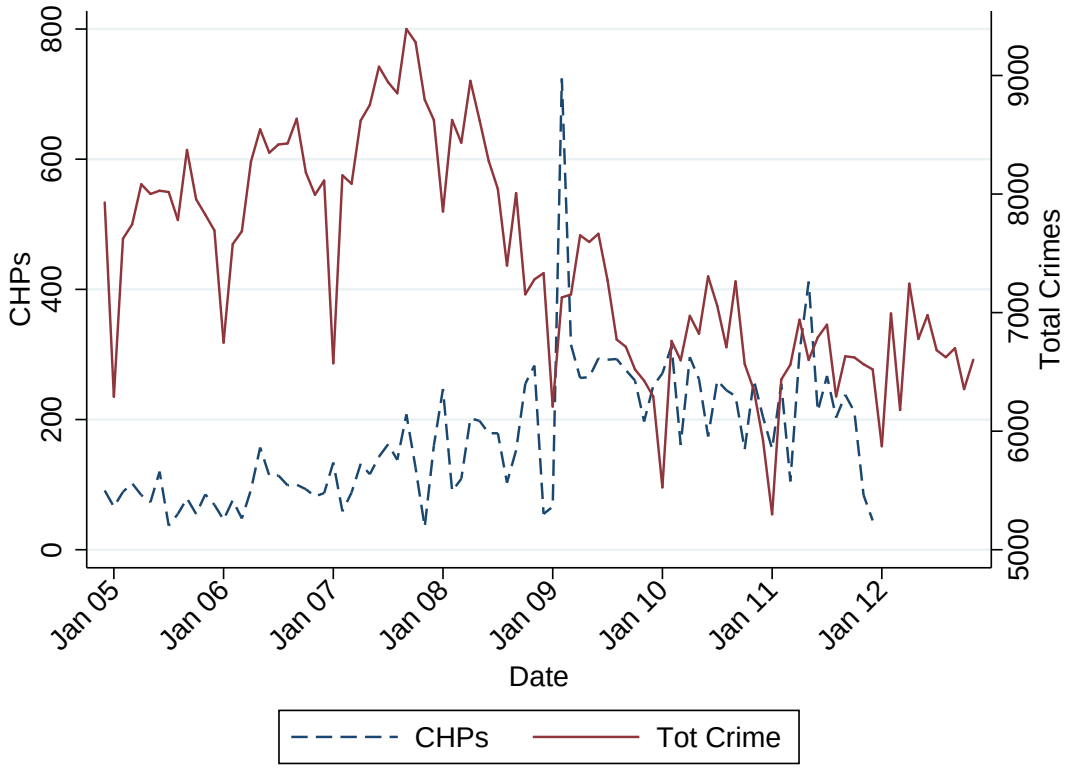
Figure 4: New Concealed Handgun Permit Holders & Total Crime



This figure provides the frequency of new CHPs issued and total crime from 2005-2013.

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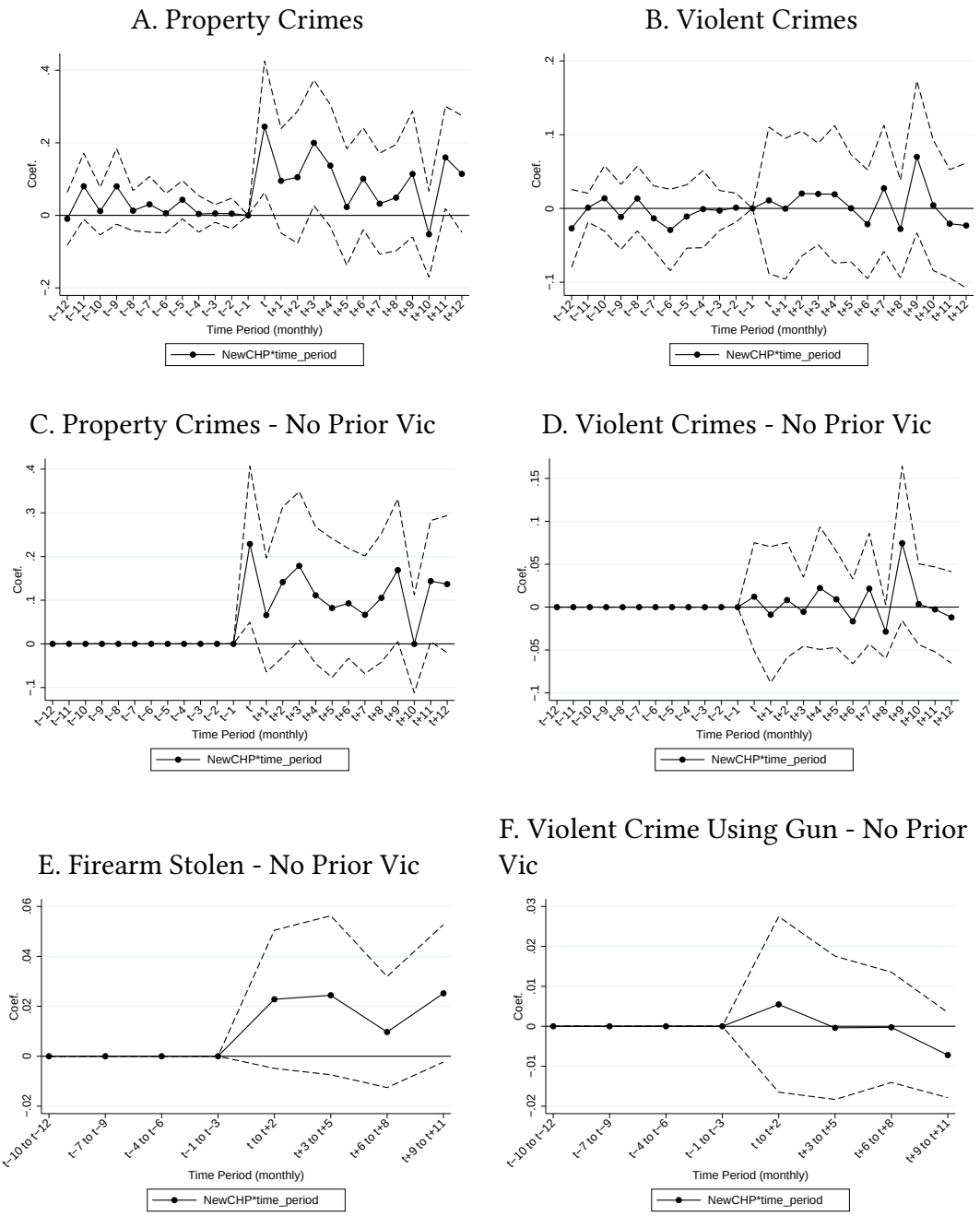
Figure 5: Temporal Impacts of New CHPs on Crime Victimization



This figure is based on Equation 1 and in panels A and B focuses on property and violent crimes for all CHP and matched non-CHP holders. Panels C and D limit results to only those individuals who were not previous crime victims at the timing a CHP was issued. Finally, Panels E and F look at gun specific crime categories of victimization for individuals who were not previous crime victims and aggregate to quarters. All models include standard errors clustered at the level of the matched group as well as the census block group by month.

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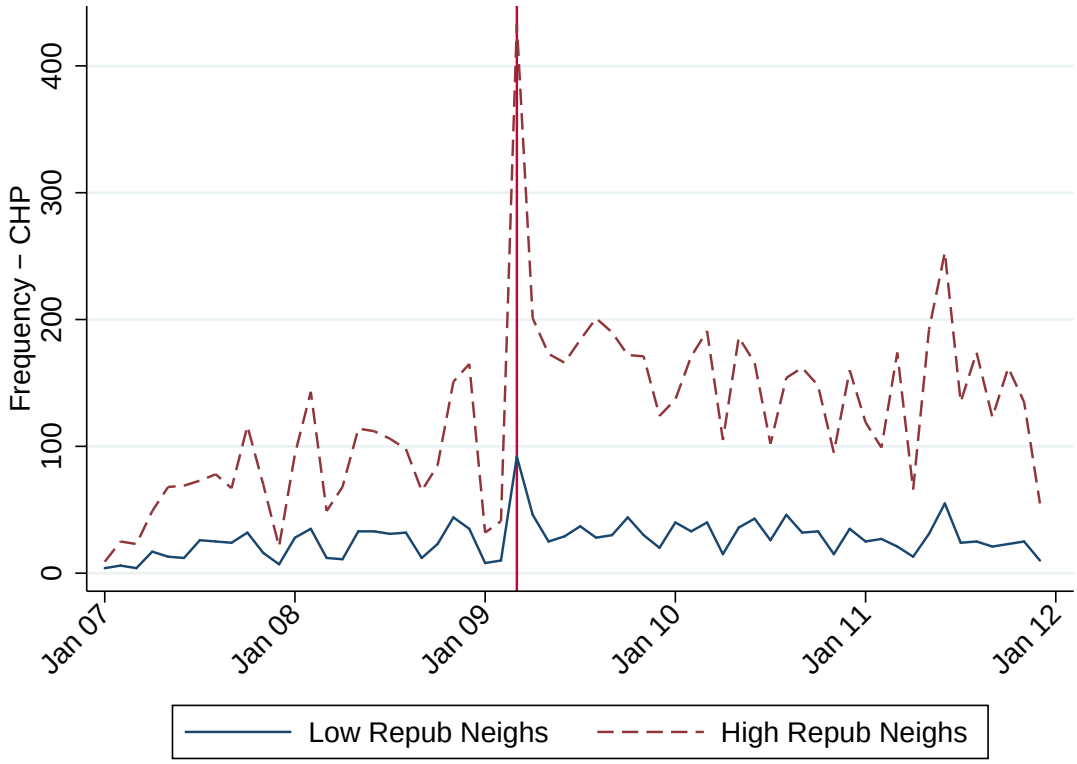
Figure 6: New CHP - Republican vs. non-Republican Neighborhoods



This figure provides the frequency of new CHPs differentially for high/low portion of residents that are Republicans within a CBG 2000 neighborhood for our main time period of study. I define High vs Low Republican neighborhoods based on the median value of share of Republicans in a neighborhood. Neighborhoods defined as Census Block Group 2000 (n=373).

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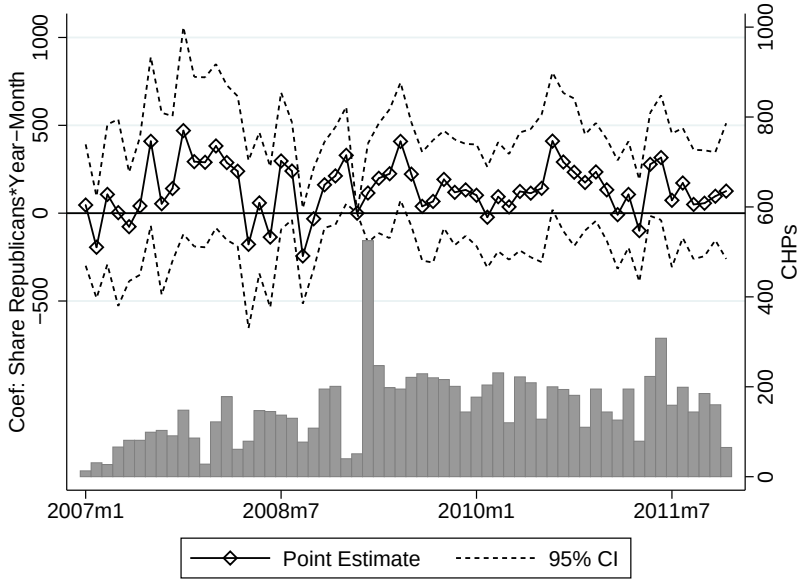
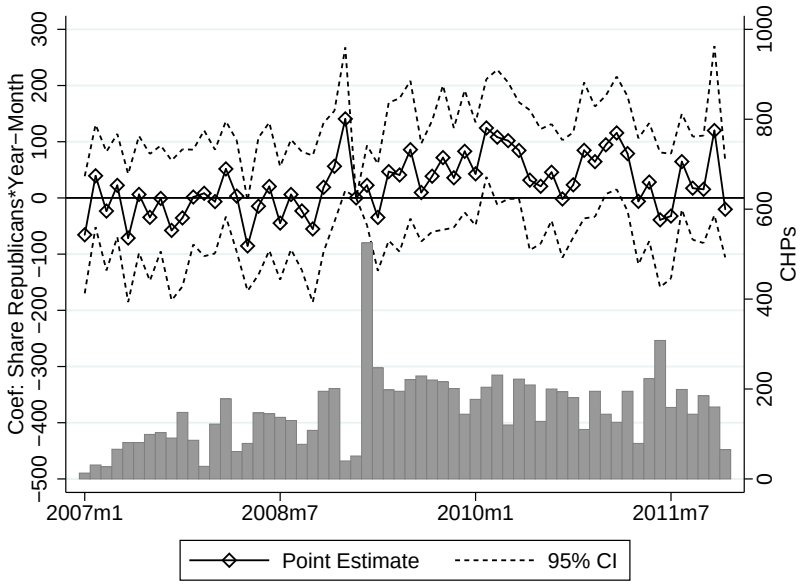
Figure 7: Pretrends in Property or Violent Crimes?

A. Property Crimes



B. Violent Crimes



These figures provide the estimates for β_1 from Equation 5 and indicate the interaction between $\mathtt{ShareRepub}_{n,t=2005}$ and a dummy for each month of the study period. I also provide a frequency histrogram of new CHPs for each month on the bottom of each figure to highlight the higher levels of CHPs including and after the election of Barack Obama.

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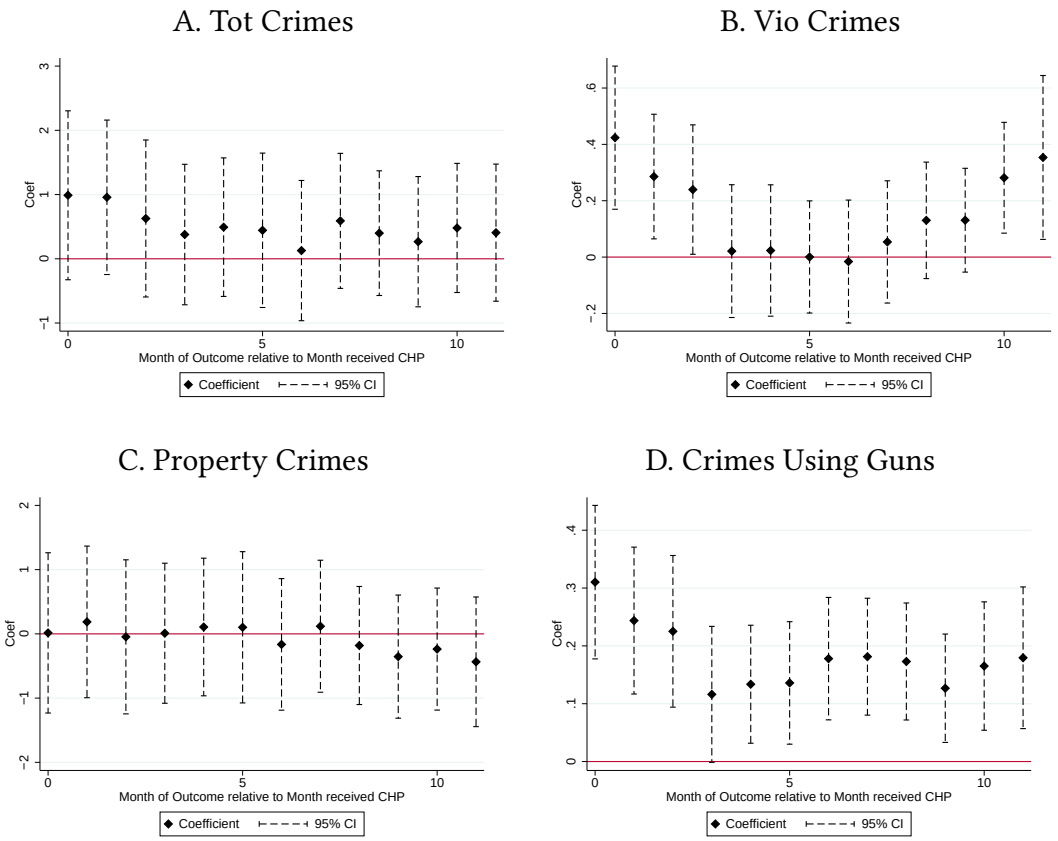
Figure 8: Temporal Impacts of New CHPs on Neighborhood Crime



This figure provides the coefficients for separate IV estimates of $Y_{n[t=k]} = \alpha + \beta_1 \widehat{CHP_{n,[t,t-1]}} + \delta_t + \gamma_n + \epsilon_{nt}$ where k is defined on the x-axis. I define violent crimes as arson, (aggravated and simple) assault, manslaughter, rape, robbery and define property crimes as auto theft, burglary, forgery/fraud , hit and run, larceny, stolen property and vandalism. Crimes using guns include any violent crime for which a weapon was mentioned in the crime report. All models include standard errors clustered by neighborhood and also at the month by initial neighborhood share of Republicans (binning shares into 20 equally sized groups) level. This generates 1,200 (20*60) clusters given the 60 months of data.

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Table 1: Annual Summary Statistics

	Registered Voters - No CHP			Registered Voters - w/ CHP		
	Mean	*Std Dev*	*N*	*Mean*	*Std Dev*	*N*
Permit Variables						
CHP Issued (year t)	0.00	(0.00)	4,070,230	19.81	(39.85)	68,290
New CHP Issued (year t)	0.00	(0.00)	4,070,230	12.44	(33.01)	68,290
Individual Attributes						
Homebuyer's Income ($ 10k)	10.09	(19.15)	962,934	11.90	(26.62)	17,747
Democrat	0.45	(0.50)	4,070,230	0.27	(0.45)	68,290
Republican	0.27	(0.45)	4,070,230	0.48	(0.50)	68,290
Black	0.30	(0.46)	4,070,230	0.18	(0.38)	68,290
Hispanic	0.02	(0.15)	4,070,230	0.01	(0.10)	68,290
Age 30-50	0.47	(0.50)	4,070,230	0.47	(0.50)	68,290
Age 51+	0.29	(0.45)	4,070,230	0.41	(0.49)	68,290
Registered in NC > 5 years	0.49	(0.50)	4,070,230	0.69	(0.46)	68,290
Born Out-of-State	0.69	(0.46)	3,151,464	0.68	(0.47)	52,587
Crime Victim						
Victim - Prop Crime (year t)	0.03	(0.16)	4,070,230	0.05	(0.21)	68,290
Victim - Vio Crime (year t)	0.01	(0.09)	4,070,230	0.01	(0.08)	68,290
Victim - Tot Crime (year t)	0.04	(0.23)	4,070,230	0.06	(0.26)	68,290
Neigh Crime						
Prop Crimes in Neigh (year t)	202	(234)	4,070,230	171	(217)	68,290
Vio Crimes in Neigh (year t)	48	(52)	4,070,230	37	(46)	68,290
Tot Crimes in Neigh (year t)	306	(333)	4,070,230	253	(307)	68,290

Summary Statistics are based on a annual panel of individual registered voters from 2007-2011 merged with concealed handgun permits (CHP) in Mecklenburg County (Charlotte), NC. Criminal justice data is based on individual reported crimes and arrests from 2005-2013. Datasets are merged primarily based on full name, birth year and address. Neigh is defined as Census Block Group 2000 in all models unless otherwise stated.

Permit variables for individuals in rows 1 and 2 are scaled to be 0 or 100 and can be interpreted as percentage points. I define violent crimes as arson, (aggravated and simple) assault, manslaughter, rape, robbery and define property crimes as auto theft, burglary, forgery/fraud , hit and run, larceny, stolen property and vandalism.

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Table 2: Individual Crime Victimization and CHPs

	(1) Any Crime Victim	(2) Vio Crime Victim	(3) Prop Crime Victim	(4) Serious Injury Victim	(5) Gun Stolen Victim	(6) Vio Crime Using Gun Victim
A) All						
New CHP Holder*Post	0.0976***	0.0141	0.0783***	−0.0053	0.0262***	0.0034
	(0.0282)	(0.0100)	(0.0245)	(0.0034)	(0.0077)	(0.0048)
New CHP Holder	0.0173*	−0.0077**	0.0224**	0.0000	0.0024	−0.0008
	(0.0103)	(0.0033)	(0.0093)	(0.0000)	(0.0022)	(0.0016)
Dep. Var. (mean) p.p.	0.271	0.038	0.206	0.005	0.014	0.010
Observations	941,046	941,046	941,046	941,046	941,046	941,046
B) No Prior Victims						
New CHP Holder*Post	0.1375***	0.0070	0.1176***	−0.0014	0.0228***	−0.0003
	(0.0250)	(0.0089)	(0.0218)	(0.0030)	(0.0071)	(0.0041)
Dep. Var. (mean) p.p.	0.202	0.025	0.154	0.006	0.010	0.006
Observations	849,848	849,848	849,848	849,848	849,848	849,848
C) Prior victims only						
New CHP Holder*Post	0.1997	0.1128	0.0809	−0.0123	0.0937*	0.0444
	(0.2168)	(0.0818)	(0.1903)	(0.0135)	(0.0564)	(0.0413)
New CHP Holder	0.0605	−0.0877*	0.1506	−0.0000	0.0142	−0.0095
	(0.1346)	(0.0487)	(0.1233)	(0.0000)	(0.0271)	(0.0245)
Dep. Var. (mean) p.p.	1.238	0.192	0.913	0.017	0.067	0.061
Observations	91,198	91,198	91,198	91,198	91,198	91,198

Dataset is based on a monthly panel of individual registered voters from 2007-2011 merged with each concealed handgun permits (CHP) in Mecklenburg County (Charlotte), NC. Criminal justice data is based on individual reported crimes and arrests from 2005-2013. Datasets are merged primarily based on full name, birth year and address. Neigh is defined as Census Block Group 2000 in all models unless otherwise stated.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Difference-in-difference model based on pre-CHP matched individuals on observables and specified as Equation 2. Since I define panels B and C based on previous values of the dependent variable, I calculate the mean of the dependent variable based only on the post CHP time period for these two panels. I define violent crimes as arson, (aggravated and simple) assault, manslaughter, rape, robbery and define property crimes as auto theft, burglary, forgery/fraud , hit and run, larceny, stolen property and vandalism. Standard errors are clustered by matching group as well as by census block group by month.

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Table 3: Individual Arrest and CHPs

	(1) Any Arrest	(2) Vio Crime Arrest	(3) Prop Crime Arrest	(4) Weapons Arrest	(5) Other Arrest	(6) Felony Arrest
A) All						
New CHP Holder*Post	−0.0167*	−0.0014	−0.0075**	−0.0014	−0.0069	−0.0020
	(0.0093)	(0.0036)	(0.0032)	(0.0019)	(0.0075)	(0.0020)
New CHP Holder	−0.0032	−0.0018	−0.0016	0.0022	−0.0020	−0.0002
	(0.0046)	(0.0016)	(0.0014)	(0.0016)	(0.0037)	(0.0014)
Dep. Var. (mean) p.p.	0.034	0.006	0.006	0.001	0.021	0.001
Observations	1,155,336	1,155,336	1,155,336	1,155,336	1,155,336	1,155,336
B) No prior arrests						
New CHP Holder*Post	−0.0114	−0.0014	−0.0042*	0.0007	−0.0070	−0.0008
	(0.0078)	(0.0031)	(0.0022)	(0.0010)	(0.0065)	(0.0011)
Dep. Var. (mean) p.p.	0.042	0.008	0.007	0.001	0.027	0.002
Observations	1,148,124	1,148,124	1,148,124	1,148,124	1,148,124	1,148,124
C) Prior arrests only						
New CHP Holder*Post	0.8522	0.3446	−0.7855	−0.2068	1.4999*	0.0321
	(1.2449)	(0.4438)	(0.7263)	(0.2619)	(0.8754)	(0.3429)
New CHP Holder	−2.1683**	−0.3496	−0.4046	0.2098	−1.6239**	−0.2128
	(0.9898)	(0.3190)	(0.4704)	(0.2657)	(0.7709)	(0.2657)
Dep. Var. (mean) p.p.	1.305	0.240	0.320	0.027	0.719	0.027
Observations	7,212	7,212	7,212	7,212	7,212	7,212

Dataset is based on a monthly panel of individual registered voters from 2007-2011 merged with each concealed handgun permits (CHP) in Mecklenburg County (Charlotte), NC. Criminal justice data is based on individual reported crimes and arrests from 2005-2013. Datasets are merged primarily based on full name, birth year and address. Neigh is defined as Census Block Group 2000 in all models unless otherwise stated.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Difference-in-difference model based on pre-CHP matched individuals on observables and specified as Equation 2. Since I define panels B and C based on previous values of the dependent variable, I calculate the mean of the dependent variable based only on the post CHP time period for these two panels. All models include standard errors clustered at the level of the matched group as well as the census block group by month.

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Table 4: Neigh CHP and Crime - Main Models

	(1) Tot Crime	(2) Vio Crime	(3) Prop Crime	(4) Serious Injury Victim	(5) Crimes w Gun Stolen	(6) Crimes Using Gun
A) OLS , CBG FEs - *Crime (t=0 to t=2)*						
New CHPs	0.053	0.020	0.016	0.005	0.001	0.026**
	(0.157)	(0.028)	(0.134)	(0.007)	(0.012)	(0.012)
B) IV - *Crime (t=0 to t=2)*						
New CHPs	4.157**	0.973***	1.876	0.167*	0.253*	0.850***
	(1.861)	(0.317)	(1.809)	(0.087)	(0.150)	(0.199)
Cragg-Donald Wald 1st Stage F-Stat	397	397	397	397	397	397
Dep. Var. (mean)	59.07	9.99	37.49	0.79	1.33	2.06
Observations	22,260	22,260	22,260	22,260	22,260	22,260
C) IV - *Crime (t=3 to t=5)*						
New CHPs	2.100	−0.005	1.127	0.081	0.255*	0.448***
	(1.533)	(0.295)	(1.495)	(0.072)	(0.147)	(0.159)
Cragg-Donald Wald 1st Stage F-Stat	397	397	397	397	397	397
Dep. Var. (mean)	58.55	9.98	36.92	0.75	1.25	2.01
Observations	22,260	22,260	22,260	22,260	22,260	22,260
D) IV - *Crime (t=0 to t=5)*						
New CHPs	6.258*	0.968*	3.003	0.248*	0.507*	1.299***
	(3.236)	(0.553)	(3.144)	(0.146)	(0.286)	(0.345)
Cragg-Donald Wald 1st Stage F-Stat	397	397	397	397	397	397
Dep. Var. (mean)	117.61	19.97	74.41	1.55	2.58	4.07
Observations	22,260	22,260	22,260	22,260	22,260	22,260

* p < 0.1, ** p < 0.05, *** p < 0.01. The main sample for this table is a monthly panel of Census Block Groups 2000 neighborhoods from 2007-2011. All models include annual covariates for neighborhood based on lagged annual voter registration data. I define violent crimes as arson, (aggravated and simple) assault, manslaughter, rape, robbery and define property crimes as auto theft, burglary, forgery/fraud , hit and run, larceny, stolen property and vandalism.

The mean (st. dev) of the independent variable $CHP_{n,[t,t-1]}$ is 0.8 (1.5).

All models include standard errors clustered by neighborhood and also at the month by initial neighborhood share of Republicans (binning shares into 20 equally sized groups) level. This generates 1,200 (20*60) clusters given the 60 months of data

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Table 5: Neigh CHP and Composition of Crimes

	Serious Victim Injury %	Prop Crimes w Gun Stolen %	Vio Crimes Using Gun %	Vio Crimes using Non-Gun Weapon %	Tot Crimes Cleared %	Gun Crimes Cleared %
IV - *Crime (t=0 to t=5)*						
New CHPs	0.018	−0.003	0.035***	0.000	−0.001	0.006
	(0.014)	(0.010)	(0.012)	(0.003)	(0.009)	(0.014)
1st Stage F-Stat	459	535	459	518	518	418
Dep. Var. (mean)	0.119	0.036	0.178	0.021	0.197	0.293
Observations	19,777	20,934	19,777	21,325	21,325	12,772

* p < 0.1, ** p < 0.05, *** p < 0.01. The main sample for this table is a monthly panel of Census Block Groups 2000 neighborhoods from 2007-2011. All models include annual covariates for neighborhood based on lagged annual voter registration data. Observation counts vary between model since some CBG by month observations have zero for total, property, violent or gun-related (gun used in crime or stolen) crimes thus making the denominator of the dependent variable in columns headings to be undefined and treated as missing in these models.

All models include standard errors clustered by neighborhood and also at the month by initial neighborhood share of Republicans (binning shares into 20 equally sized groups) level. This generates 1,200 (20*60) clusters given the 60 months of data

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Table 6: Neigh CHP and Additional Gun Crimes

	Gun Stolen Crimes #	Multiple Guns Thefts #	Gun Stolen from Car Crimes	Gun Stolen from Home Crimes	Gun Stolen from Other Crimes	Tot. Prop Val $000s Stolen	Deaths Using Gun
IV - *Crime (t=0 to t=5)*							
New CHPs	0.580**	−0.012	0.109	0.131	0.341***	17.307	0.023
	(0.263)	(0.071)	(0.114)	(0.140)	(0.128)	(20.000)	(0.016)
Cragg-Donald Wald 1st Stage F-Stat	465	465	465	465	465	465	465
Dep. Var. (mean)	2.58	.455	.415	1.31	1.00	128	.0488
Observations	22,260	22,260	22,260	22,260	22,260	22,260	22,260

* p < 0.1, ** p < 0.05, *** p < 0.01. The main sample for this table is a monthly panel of Census Block Groups 2000 neighborhoods from 2007-2011. All models include annual covariates for neighborhood based on lagged annual voter registration data. Multiple gun thefts represent the counts of reported stolen firearm crimes in which multiple guns were reported stolen.

All models include standard errors clustered by neighborhood and also at the month by initial neighborhood share of Republicans (binning shares into 20 equally sized groups) level. This generates 1,200 (20*60) clusters given the 60 months of data

Table 7: Social Costs of Gun Ownership

Crime Category	Mean Ann. Neigh Tot Crime	Coef.	Total Ann Inc. in Crime for additional CHP	Costs Estimates McCollister et al. (2010)	Neigh Social Cost for add. CHP	Per Capita Social Cost of add. CHP
Assaults	32.0	0.395	0.79	19,537	$ 15,434	
Auto Theft	10.7	1.818***	3.64	10,534	$ 38,302	
Burglary	25.8	0.020	0.04	6,170	$ 247	
Drugs	13.6	0.783**	1.57	8,347	$ 13,071	
Forgery	12.1	1.024***	2.05	5,265	$ 10,783	
Hit and Run	10.0	0.486**	0.97	8,347	$ 8,113	
Larceny	35.7	1.242	2.48	3,523	$ 8,751	
Murder	0.1	0.017	0.03	1,278,424	$ 43,466	
Other	30.7	1.030*	2.06	8,347	$ 17,195	
Robbery	6.0	0.836***	1.67	21,398	$ 35,777	
Sex Offenses	1.4	0.087	0.17	41,247	$ 7,177	
Stolen Property	0.9	-0.033	-0.07	7,974	$ (526)	
Theft from Auto	34.0	-1.557	-3.11	3,523	$ (10,971)	
Vandalism	19.6	0.461	0.92	4,860	$ 4,481	**$ 76.55**
Total	232.5		13.2		$ 191,301	

(1) In order to generate annual impacts, I use the fact that CHPs are measured for two months and that outcomes are measured as 6 months outcomes. Therefore, each CHP generates two 6 month crime effects and thus doubling the coefficient captures the total effects. I assume no additional crime greater than 6 months in the future and simply report total effects for the year.

(2) If cost estimates from MFF (2010) are not available , I use the weighted average of social costs across all crime categories with costs ($8,347) estimates. Given the time period of analysis and cost estimates, costs are roughly based on 2010 ($s)

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A. Online Appendix

A.1. Master Dataset Creation - Individuals

This section provides some more details on the merging on datasets using individual information. The base dataset for the creation of the master dataset used in all analysis is annual snapshots of voter registration records from the state of North Carolina for 2005 through 2013 (https://dl.ncsbe.gov/index.html?prefix=data/). The main dataset that provides individual level data on Concealed Handgun Permit (CHP) holders is from the NC Department of Public Safety. This data was requested in 2012 and contains information on full name, residential address and date of birth. In 2014, state senate bill 226 removed the public record element of gun permit data and thus this data is no longer available for public request and thus my analysis is limited to the data previously requested that contained 2007 through the end of 2011. Additional datasets include detailed property transaction and assessor records from Mecklenburg County, NC. I also have merged these property records with HMDA based data that contains information on homebuyer mortgage states income. A number of scholars have incorporated HMDA public data to provide individual homebuyer income and race for new residents to a specific property and neighborhood based on loan amount, census tract of the property and lender name. This information is limited to mortgaged properties. The final main dataset merged with voter registration records is incident level data for reported crimes in Charlotte/Mecklenburg County for 2005 through 2013. Some unique elements of this crime dataset include identifying individual data on crime victimization, lots of specific details on the location and crime categories of a reported crime, date and time of incident.

Voter registration data is merged with CHP data primarily based on full name, birth year and address. In some cases, data is merged based on full names and birth year that are unique in the voter registration dataset. For a small percentage of cases where voter information was missing in a given year, I imputed data from previous years forward. The merging of CHP data to voter registration data was conducted in three stages. First, I match each CHP to all unique voters based on full name and home address. This matching algorithm does a good job of linking up individuals with 91% of matcheable CHPs merged using this method. Second, I match each CHP to unique voters based on age, full name and address. This method matches about 8% of matcheable CHPs and deals with multigenerational families such as fathers and sons of the same name living together. Third, I attempted to match based on full name and age or fuzzy versions of the first two matching algorithms. These fuzzy merges provided less than 1% of matcheable observations. In the end, approximately 86% of CHPs can be directly merged to a registered voter based on full name, age and address. Registered voters in Mecklenburg County, NC represent about 75% of the population aged 18 and above. I lose approximately 4% of the original voter registration data due to missing or incomplete geographical information. Crime victimization data is merged to registered voters based on full name, birth year and address or unique full name and birth year. 0.95% of all crime victims were or became CHP holders in 2007 through 2011. This compares to a general population permit rate of 1.22%. In general there is not a metric to determine the quality of matching across these datasets since they do not contain the same variables other than match variables.

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A.2. Neighborhood Crime, Moving and Permitting

In order to better understand the relationship between neighborhood crime risk, CHP and residential moves, I estimate an annual panel model of registered voters for two outcomes: moving between last year $t-1$ and this year t and getting a CHP in year t.[50] I test the role of violent and property neighborhood crime rates since both likely effect fear of crime but vary in saliency as well as frequency.

$$y_{int} = \alpha + \beta_1 \mathrm{NeighVioCrimes}_{nt-1} + \beta_2 \mathrm{NeighPropCrimes}_{nt-1} + \beta_2 X_i t + \gamma_n + \delta_t + \epsilon_{int} \tag{6}$$

Specifically, I estimate Equation 6 for moving and then for obtaining a new permit and even include individual fixed effects in some models.[51] X_{it} provides individual covariates (given in Table 1) of which some vary over time. $\mathrm{NeighVioCrimes}_{nt-1}$ and $\mathrm{NeighPropCrimes}_{nt-1}$ provide counts of neighborhood n violent and property crimes in the previous year. Models with neighborhood fixed effects are identified off of variation in neighborhood crime levels for two similar individuals that live in the same neighborhood but in different years. Adding individual fixed effects to this model involves estimating how temporal variation in neighborhood crime relates to the timing of moving or obtaining a permit for an individual.[52] All dependent variables are scaled to be measured in percentage points (i.e. moving rates are 3.997% a year in column 2)

Appendix Table A3 directly estimates Equation 6 in columns 1 through 3 for different fixed effects and dropping imputed voter locations and inactive voters. The first results from these first three columns is that recent neighborhood violent crime makes an individual more likely to move. Specifically, column 3 estimates that a one standard deviation (52) increase in neighborhood violent crimes generates a 2.12 (0.407 * 5.2) percentage point increase in moving which is a 53% increase over mean annual move rates. For neighborhood property crimes, effects are quite imprecise and even with the higher levels of property crime, a one standard deviation increase (234) translates into 0.84 (0.036* 23.4) percentage point increase or a 21% increase over the baseline rate.

Conversely, the impacts of neighborhood crime on permits is quite limited. Columns 4 through 6 present models with neighborhood and individual fixed effects and in this case results hold across all three models. Columns 4 through 6 highlight that neighborhood crime has no almost impact on new permits. Moving into a new neighborhood did increase obtaining new permits by 12% (0.064/0.52). This result highlights that experiencing recent neighborhood crime is simply causing people to move rather than to get CHPs. This

[50] Since one may be concerned if individuals update voter registration records when they move, I provide specifications where I vary the use of imputed voter info as well as inactive voters (designated by state officials to identify deceased and voters who left the state). In years when a voter is missing from the annual snapshot, I impute voter information by simply carrying the previous year voter information forward a year.

[51] Since an individual can only obtain a new permit one time in my dataset, all annual observations after an individual obtains a new permit are dropped for that individual in estimation of models with new CHP as the outcome.

[52] The relatively short time period of this analysis (5 years) does limit the ability of substantial neighborhood change (e.g. gentrification) to influence these crime trends.

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result is consistent with a story of individuals selectively moving away from higher crime neighborhoods and getting CHPs in their new neighborhoods creating attenuation bias in OLS estimates of CHPs and neighborhood crime.

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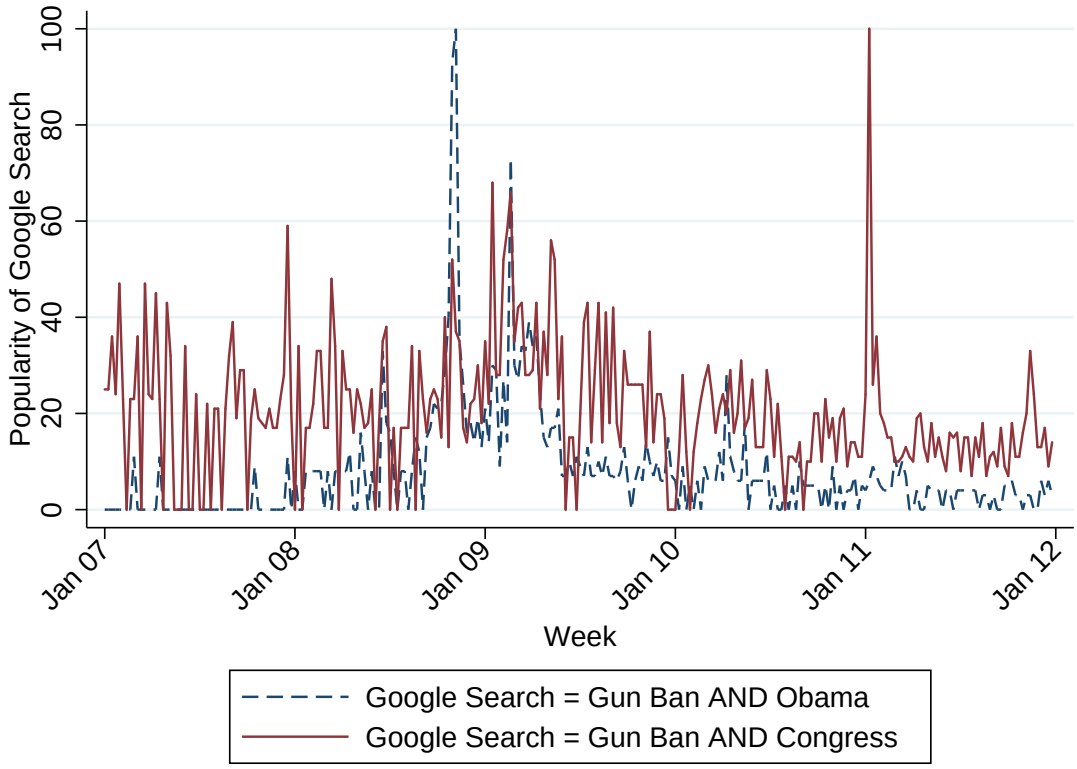
A.3. Appendix Tables and Figures

Figure A1: Google Trends around Obama Spike and Giffords Shooting



This figure provides the Google trends figure for my study period for the terms "Gun Ban AND Obama" and "Gun Ban AND Congress". The y-scale is normalized to 100 for the most popular week upon which this search terms was used in the US.

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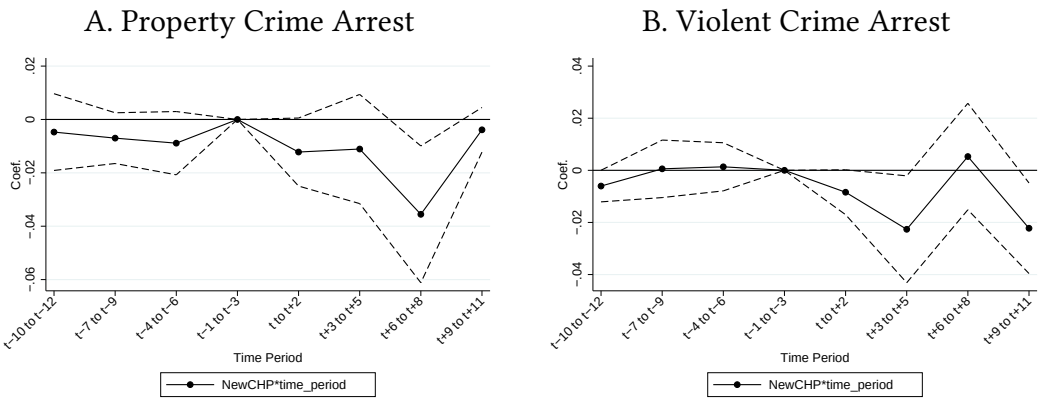
Figure A2: Temporal Impacts of New CHPs on Arrest



This figure is based on Equation 1 and arrest outcomes. Results focuse on months aggregated to quarters given the limited number of arrests for CHP holders in this dataset. All models include standard errors clustered at the matching group and neighborhood (CBG) by month.

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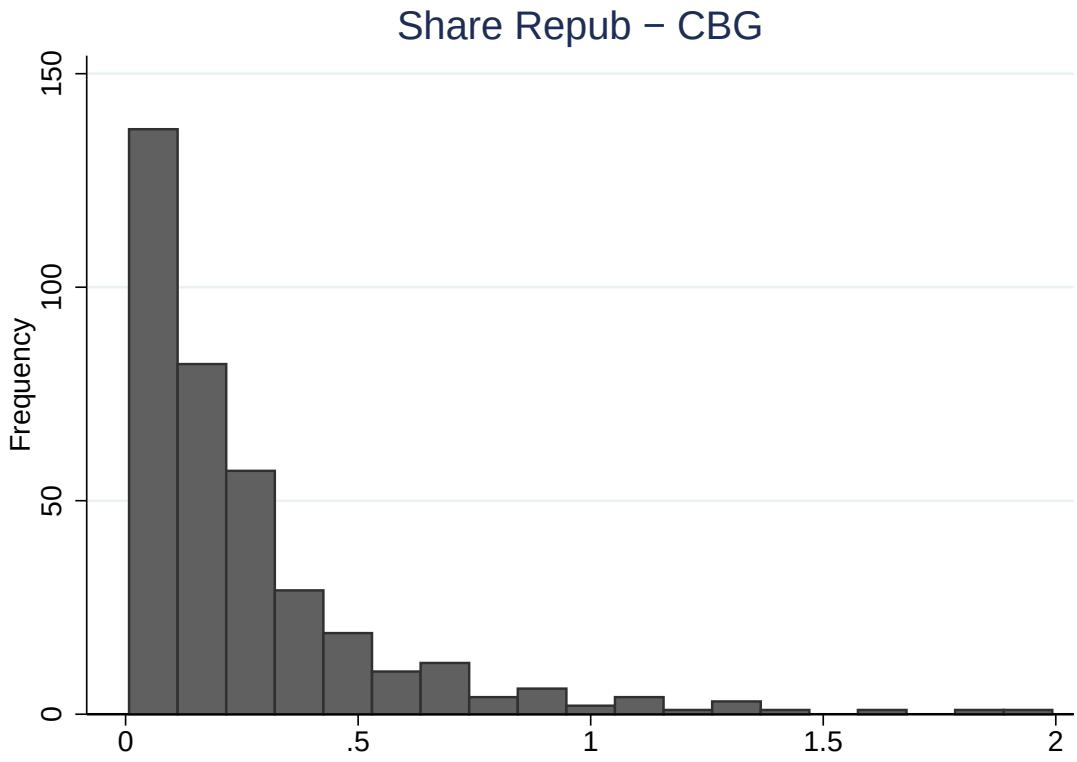
Figure A3: Distribution of Share Republicans



This figure provides the distribution of Republicans across my definition of neighborhood. The mass of this distribution equals one by construction since we define share Republican as the portion of all Republicans in our study area that lived in a given neighborhood in 2005. Neighborhoods defined as Census Block Group 2000 (n=373).

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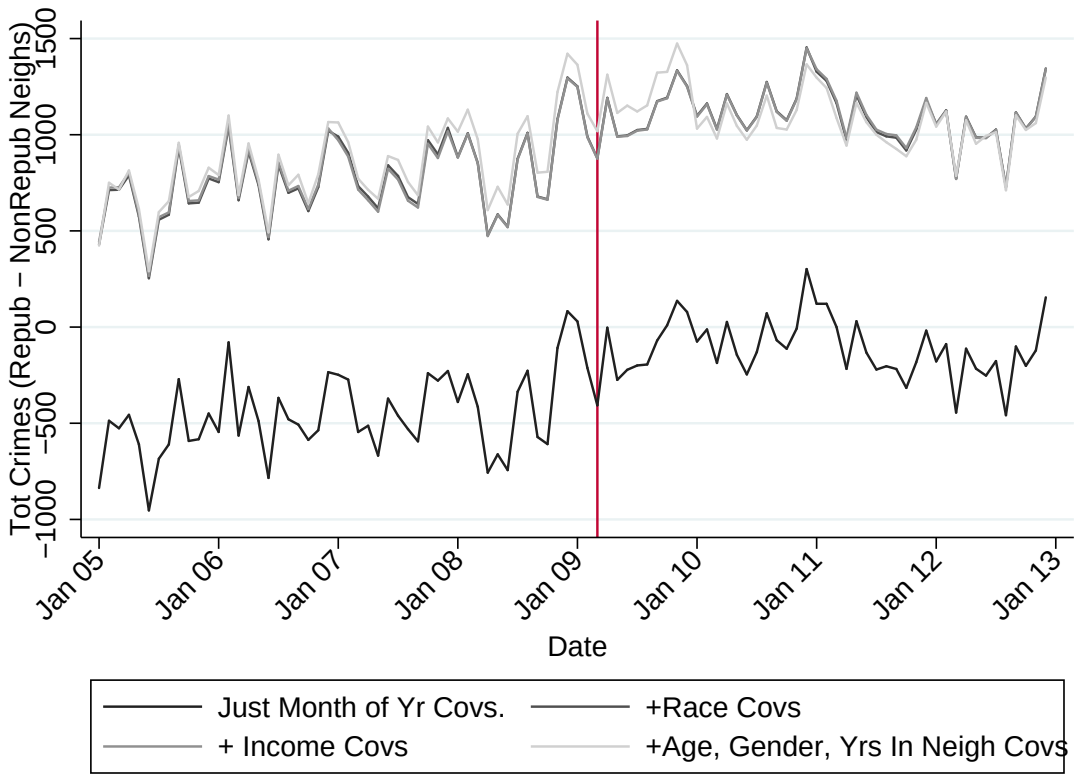
Figure A4: Total Crime - Differences b/t Republican vs. Non-Republican Neighborhoods with Different Controls



This figure provides the difference in monthly crime for above average Republican and below average Republican neighborhoods conditional on different sets of covariates. Each line adds additional controls (as specified in the legend) as it moves from darker to lighter coloring. Vertical line represents large increase in CHP right after the inauguration of Barack Obama in 2009. Neighborhoods defined as Census Block Group 2000 (n=373).

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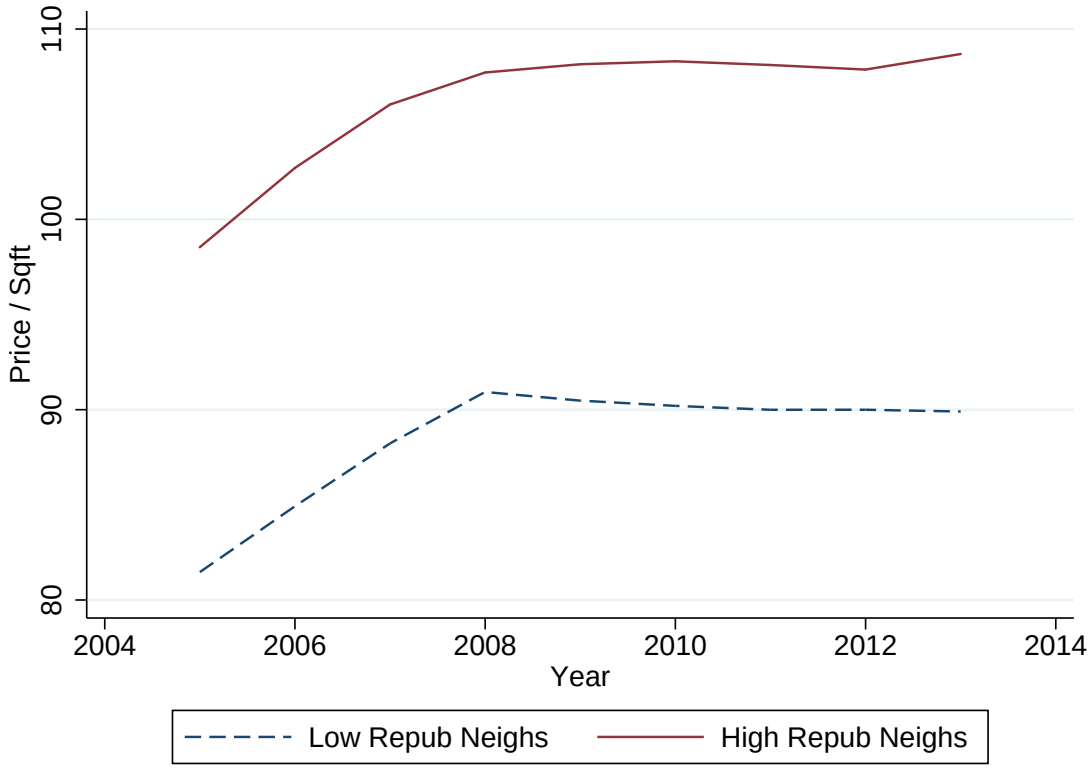
Figure A5: Neighborhood Convergence or Divergence? - Prices



This figure provides annual averages for transacted sales prices per square foot for residential properties split for above median vs below median Republican neighborhoods. I define a neighborhood as Republican in 2005 before the beginning of my study period.

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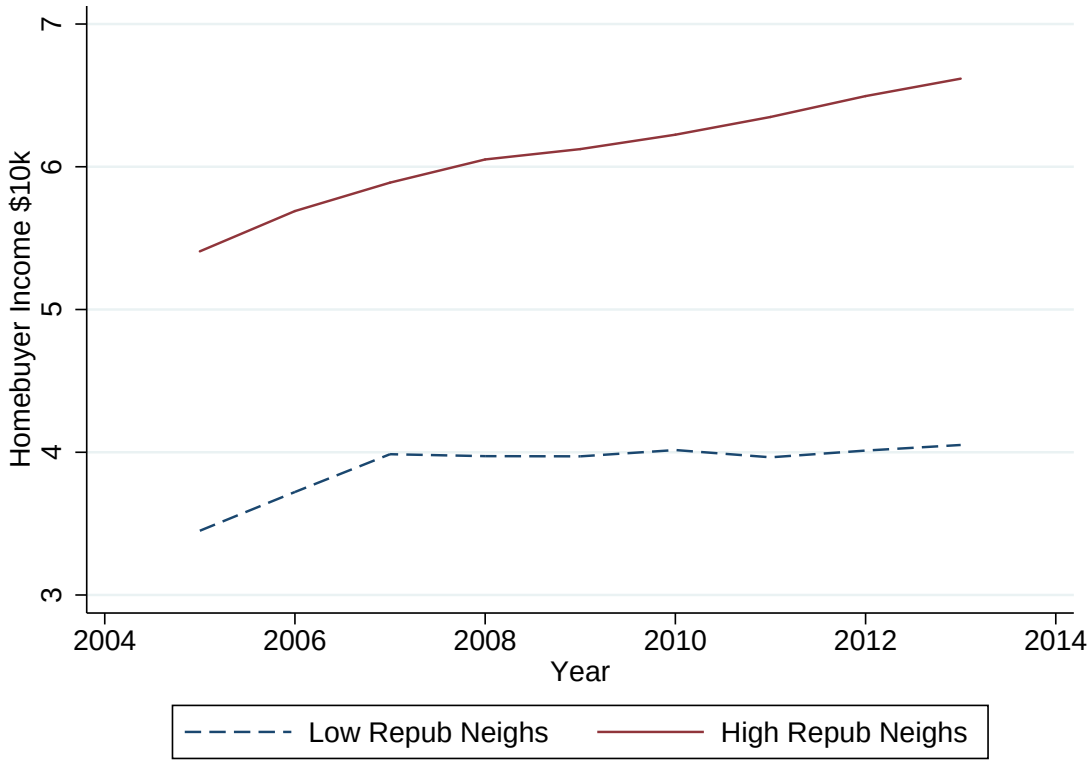
Figure A6: Neighborhood Convergence or Divergence? - Incomes of Homebuyers



This figure provides annual averages for homebuyer income based on HMDA mortgage data for above median vs below median Republican neighborhoods. I define a neighborhood as Republican in 2005 before the beginning of my study period.

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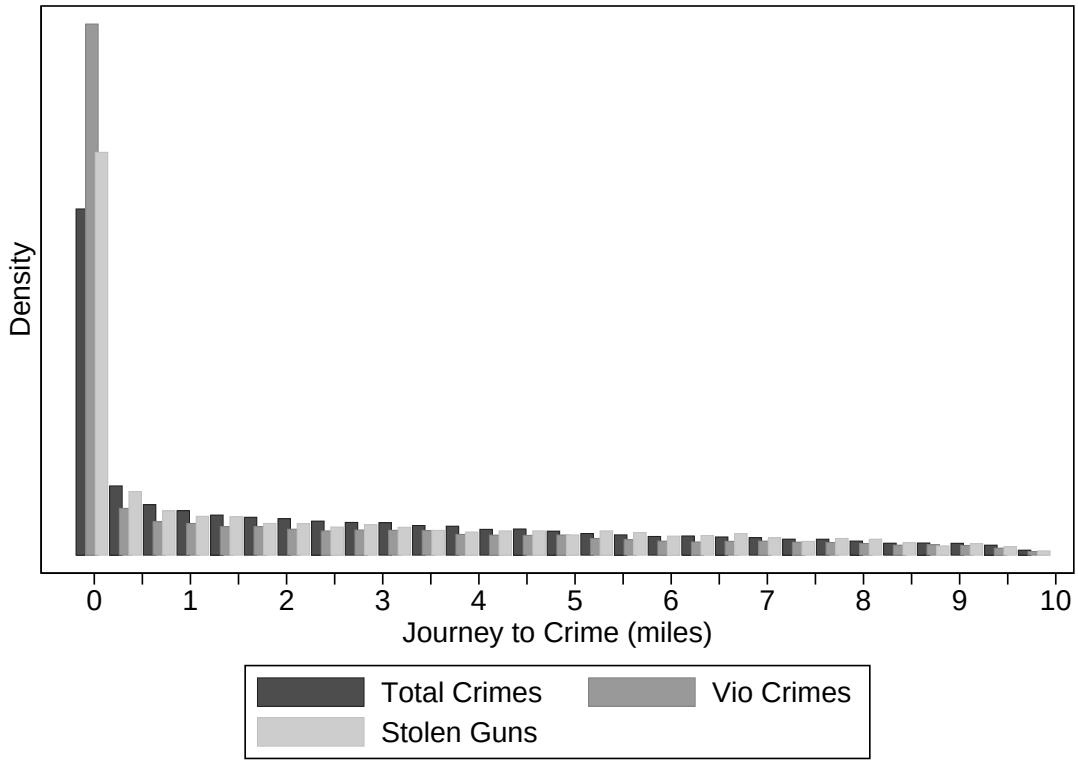
Figure A7: Journey to Crime



This figure provides the distance between an arrested individual's home address and the location of the reported crime linked to the arrest. I conduct this analysis for all reported crimes with linked arrests from 2005-2011 in Mecklenburg County, NC. This data was collected as part of a larger effort by the Charlotte Mecklenburg Police Department (CMPD) to better track crime clearance and profile potential offenders for non-cleared reported crimes.

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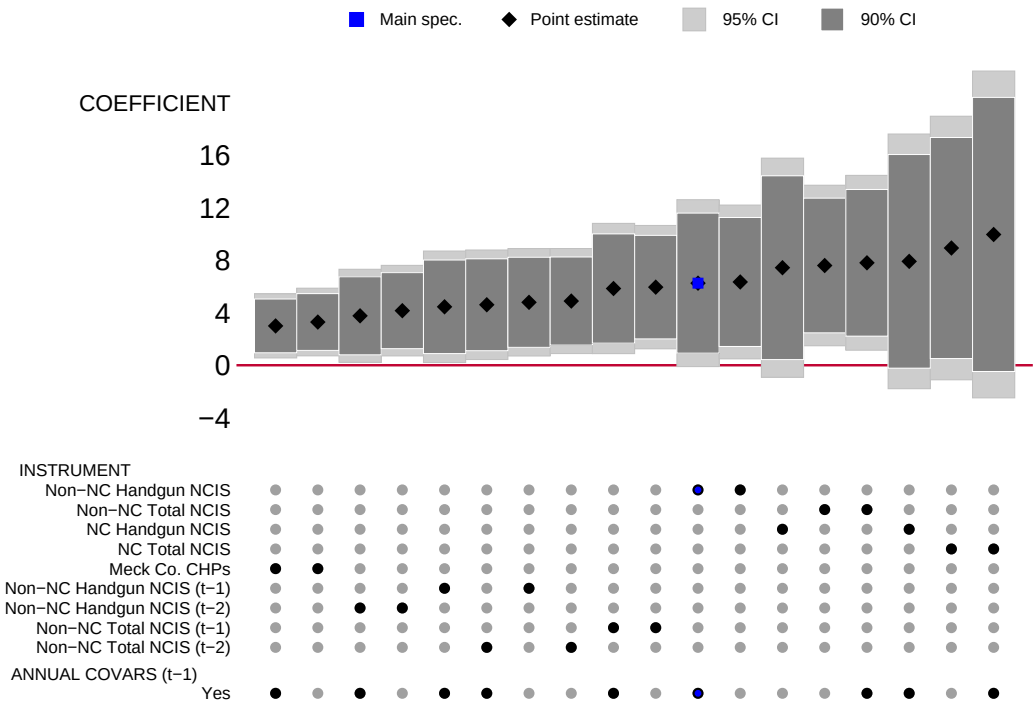
Figure A8: Specification Figure



Full Range of Different Models based on row headings. Neighborhoods defined as Census Block Group 2000 (n=373). All models include standard errors clustered by neighborhood and also at the month by initial neighborhood share of Republicans (binning shares into 20 equally sized groups) level. This generates 1,200 (20*60) clusters given the 60 months of data.

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Table A1: Explaining the Attributes of Permit Holders

	(1) Ever CHP	(2) Ever New CHP	(3) Ever New CHP *with CBG FEs*
Individual Att.			
Homebuyer's Income ($ 10k)	0.004	0.004	0.004
	(0.003)	(0.002)	(0.002)
Democrat	−0.468***	−0.289***	−0.287***
	(0.038)	(0.029)	(0.028)
Republican	0.995***	0.650***	0.644***
	(0.048)	(0.037)	(0.036)
Black	−0.070	−0.070**	−0.090***
	(0.045)	(0.033)	(0.034)
Hispanic	−0.219***	−0.183***	−0.170***
	(0.065)	(0.054)	(0.054)
age 30-50	0.415***	0.286***	0.280***
	(0.036)	(0.030)	(0.029)
age 51+	0.724***	0.273***	0.271***
	(0.053)	(0.039)	(0.038)
Registered in NC > 5 years	0.337***	−0.005	0.002
	(0.044)	(0.036)	(0.035)
Born Out-of-State	−0.292***	−0.176***	−0.167***
	(0.042)	(0.033)	(0.033)
Homebuyer's Income Missing	0.085*	0.012	0.029
	(0.050)	(0.041)	(0.041)
Neigh Vars.			
CBG Black (%)	−0.778***	−0.381***	
	(0.200)	(0.136)	
CBG Hispanic (%)	−0.620	−0.324	
	(0.496)	(0.340)	
CBG Pop Density (00s pop/sq mi)	−0.011***	−0.008***	
	(0.002)	(0.002)	
CBG Median HH Income (00s)	0.001***	0.001***	
	(0.000)	(0.000)	
HHs in Poverty (%)	−0.263**	−0.210***	
	(0.116)	(0.079)	
CBG College Grad (%)	−1.474***	−0.884***	
	(0.443)	(0.311)	
CBG HS Grad (%)	0.567	0.160	
	(0.401)	(0.290)	
Neigh Prop Crime (2006)	−0.000	0.000	
	(0.000)	(0.000)	
Neigh Vio Crime (2006)	0.000	−0.000	
	(0.001)	(0.001)	
Dep. Var. (mean)	1.65	1.04	1.04
Observations	827,759	827,759	827,759

Dataset is based on a snapshot on individual registered voters in 2007 merged with concealed handgun permits (CHP) in Mecklenburg County (Charlotte), NC from 2007-2011. Criminal justice data is based on individual reported crimes in 2006. Neigh is defined as Census Block Group 2000. Dependent variable is equal to 100 if an individual obtained a (new) CHP between 2007-2011. Homebuyer's Income is set equal to zero and the Homebuyer's Income Missing is set equal to one for individuals who did not have a homebuyer income information available from the merging of HMDA and property records. Individuals who purchased a home prior to 1998, never purchased a home or did not obtain a mortgage to purchase home would not have homebuyer income in this dataset.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

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Table A2: Neigh Spillovers

Crime Category	Number of Obs.	Same Neigh JTC	Diff Neigh JTC	Diff Neigh & Diff Levels of CHPs
Total Crimes	83,485	37.2%	62.8%	**23.3%**
Violent Crimes	25,986	52.9%	47.1%	**17.5%**
Property Crimes	24,078	21.8%	78.2%	**30.7%**
Crime w Serious Injury	1,511	43.6%	56.4%	**20.0%**
Stolen Gun Crimes	4,494	42.5%	57.5%	**21.9%**
Crime Using Gun	7,671	33.0%	67.0%	**24.6%**

JTC = Journey to Crime. I measure JTC as the distance between the reported crime location and the home address of the individual that was arrested for the reported crime. Same/Diff Neigh columns refers to the portion of crimes with linked arrestee info for which arrestee home address and reported crime address are in the same/different CBGs. Diff Neigh and Diff Levels of CHPs indicates that an arrestee is located in a different CBG than the reported crimes and these two different CBGs vary in levels of CHPs (above/below median).

Table A3: Neighborhood Sorting, Crime and Timing of CHPs

	(1) Moved Neigh (t-1) to t	(2) Moved Neigh (t-1) to t	(3) Moved Neigh (t-1) to t	(4) New Permit t	(5) New Permit t	(6) New Permit t
Neigh Vio Crime(0s) (t-1)	0.441**	0.477	0.407*	0.002	0.002	0.008
	(0.224)	(0.296)	(0.222)	(0.003)	(0.004)	(0.005)
Neigh Prop Crime(0s) (t-1)	0.016	0.025	0.036	0.001	0.001	0.001
	(0.073)	(0.087)	(0.069)	(0.001)	(0.001)	(0.001)
Moved Neigh ((t-1) to t)				0.126***	0.122***	0.064**
				(0.027)	(0.029)	(0.027)
Dep. Var. (mean)	2.546	3.997	3.997	0.328	0.521	0.521
Observations	4,136,775	2,437,579	2,437,579	4,124,803	2,425,890	2,425,890
Individual Controls & CBG FE	✓	✓	✓	✓	✓	✓
Individual FE	-	-	✓	-	-	✓
Drop Imputed/Inactive Voters	-	✓	✓	-	✓	✓

Dataset is based on a annual panel of individual registered voters from 2007-2011 merged with each concealed handgun permit (CHP) issued in Mecklenburg County (Charlotte), NC. Criminal justice data is based on individual reported crimes and arrests from 2005-2013. Neigh is defined as Census Block Group 2000 in all models unless otherwise stated.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Standard errors clustered at the level of neighborhood or individual fixed effects. For models using CHP as the dependent variable, I drop observations in the time periods after an individual obtains a CHP. All models include year fixed effects. Standard errors are clustered by individual.

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Table A4: Neigh CHP and Crime - Other Crime Categories

	Assaults	Burglaries	Drugs	Auto Theft	Larceny	Forg Fraud	Vandalism	Robbery	Weapons
IV - *Crime (t=0 to t=5)*									
New CHPs	0.395	0.020	0.783**	1.818***	1.242	1.024***	0.461	0.836***	0.279***
	(0.470)	(0.988)	(0.382)	(0.521)	(1.075)	(0.341)	(0.538)	(0.232)	(0.095)
Cragg-Donald Wald 1st Stage F-Stat	465	465	465	465	465	465	465	465	465
Dep. Var. (mean)	15.98	12.90	6.78	5.37	17.86	6.07	9.79	2.99	0.98
Observations	22,260	22,260	22,260	22,260	22,260	22,260	22,260	22,260	22,260

* p < 0.1, ** p < 0.05, *** p < 0.01. The main sample for this table is a monthly panel of Census Block Groups 2000 neighborhoods from 2007-2011. All models include annual covariates for neighborhood based on lagged annual voter registration data.

The mean (st. dev) of the independent variable $CHP_{n,[t,t-1]}$ is 0.8 (1.5).

All models include standard errors clustered by neighborhood and also at the month by initial neighborhood share of Republicans (binning shares into 20 equally sized groups) level. This generates 1,200 (20*60) clusters given the 60 months of data

Table A5: Neigh CHP and Crime - Robustness

	(1) Tot Crime	(2) Vio Crime	(3) Prop Crime	(4) Serious Injury Victim	(5) Crimes w Gun Stolen	(6) Crimes Using Gun
A) IV - *Crime (t=0 to t=11)*						
New CHPs	11.309*	1.953**	4.313	0.537*	1.396***	2.378***
	(5.860)	(0.905)	(5.542)	(0.289)	(0.529)	(0.605)
Cragg-Donald Wald 1st Stage F-Stat	397	397	397	397	397	397
Dep. Var. (mean)	232.21	39.78	145.95	2.92	4.82	7.90
Observations	22,260	22,260	22,260	22,260	22,260	22,260
B) IV - Predicted CHP Holder - *Crime (t=0 to t=5)*						
New CHPs	7.540**	1.615***	3.029	0.344**	0.699**	1.653***
	(3.777)	(0.614)	(3.654)	(0.159)	(0.322)	(0.400)
Cragg-Donald Wald 1st Stage F-Stat	420	420	420	420	420	420
Dep. Var. (mean)	117.61	19.97	74.41	1.55	2.58	4.07
Observations	22,260	22,260	22,260	22,260	22,260	22,260
C) IV that excludes Neighboring States						
New CHPs	6.166*	0.930*	2.926	0.243	0.503*	1.276***
	(3.216)	(0.548)	(3.127)	(0.147)	(0.283)	(0.338)
Cragg-Donald Wald 1st Stage F-Stat	401	401	401	401	401	401
Dep. Var. (mean)	117.61	19.97	74.41	1.55	2.58	4.07
Observations	22,260	22,260	22,260	22,260	22,260	22,260
D) IV - *Just Obama Spike & Crime (t=0 to t=2)*						
New CHPs	4.764***	0.979*	2.958**	−0.020	0.068	0.649**
	(1.801)	(0.501)	(1.278)	(0.139)	(0.150)	(0.255)
Cragg-Donald Wald 1st Stage F-Stat	97	97	97	97	97	97
Dep. Var. (mean)	59.92	10.11	38.24	0.80	1.36	2.23
Observations	4,823	4,823	4,823	4,823	4,823	4,823

* p < 0.1, ** p < 0.05, *** p < 0.01. The main sample for this table is a monthly panel of Census Block Groups 2000 neighborhoods from 2007-2011. All models include annual covariates for neighborhood based on lagged annual voter registration data.

Panel B is an alternative IV based on predicted gun owners and specifies the shift share instrument using a composite of individuals voter registration attributes that were identified as predictive of obtaining a CHP from column 3 in Appendix Table A1. Taking these attributes, I generated a predicted share of registered voters that increased in value due to having more Republicans, total voters, more older voters, fewer born out-of-state voters and fewer minority voters.

Panel C excludes the neighboring states of GA, SC, VA, TN in computing my instrument.

Panel D includes the main model but limits it to CHPs obtained 6 months before and after the Obama spike.

All models include standard errors clustered by neighborhood and also at the month by initial neighborhood share of Republicans (binning shares into 20 equally sized groups) level. This generates 1,200 (20*60) clusters given the 60 months of data

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Table A6: Neigh CHP and Crime - Main Models using Tracts

	(1) Tot Crime	(2) Vio Crime	(3) Prop Crime	(4) Serious Injury Victim	(5) Crimes w Gun Stolen	(6) Crimes Using Gun
A) OLS , CBG FEs - *Crime (t=0 to t=2)*						
New CHPs	0.125	0.005	0.064	0.004	0.017	0.045***
	(0.210)	(0.037)	(0.191)	(0.010)	(0.017)	(0.016)
B) IV - *Crime (t=0 to t=2)*						
New CHPs	9.926***	1.605**	6.769***	0.260*	0.587***	1.625***
	(3.184)	(0.666)	(2.588)	(0.135)	(0.209)	(0.400)
Cragg-Donald Wald 1st Stage F-Stat	155	155	155	155	155	155
Dep. Var. (mean)	164.04	27.92	104.08	2.23	3.67	5.70
Observations	8,580	8,580	8,580	8,580	8,580	8,580
C) IV - *Crime (t=3 to t=5)*						
New CHPs	5.204**	−0.225	4.847**	0.214**	0.693***	1.201***
	(2.513)	(0.510)	(2.204)	(0.094)	(0.209)	(0.323)
Cragg-Donald Wald 1st Stage F-Stat	155	155	155	155	155	155
Dep. Var. (mean)	162.61	27.89	102.53	2.11	3.44	5.56
Observations	8,580	8,580	8,580	8,580	8,580	8,580
D) IV - *Crime (t=0 to t=5)*						
New CHPs	15.129***	1.380	11.616**	0.474**	1.280***	2.826***
	(5.493)	(1.083)	(4.676)	(0.216)	(0.406)	(0.713)
Cragg-Donald Wald 1st Stage F-Stat	155	155	155	155	155	155
Dep. Var. (mean)	326.65	55.80	206.61	4.34	7.11	11.25
Observations	8,580	8,580	8,580	8,580	8,580	8,580

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. The main sample for this table is a monthly panel of Census Tract 2000 neighborhoods from 2007-2011. All models include annual covariates for neighborhood based on lagged annual voter registration data.

The mean (st. dev) of new monthly concealed handgun permits per neighborhood is 2.4 (3.3). I define violent crimes as arson, (aggravated and simple) assault, manslaughter, rape, robbery and define property crimes as auto theft, burglary, forgery/fraud , hit and run, larceny, stolen property and vandalism.

All models include standard errors clustered by neighborhood and also at the month by initial neighborhood share of Republicans (binning shares into 20 equally sized groups) level. This generates 1,200 (20*60) clusters given the 60 months of data

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